Data Journalism

Session 8: Designing effective visualizations

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What we are *not* doing today

Ground already covered

- The **grammar of graphics** and **ggplot2 basics** (Session 8 / IDS-25)
- **Chart types** and when to use each (FT Visual Vocabulary; Claus Wilke's Fundamentals of Data Visualization)
- **Cleveland/Tufte** principles: data-ink, proportional ink, comparison, etc. (Session 8 / IDS-25)
- Most of the science of **communicating uncertainty** (Session 3)
- **Basic ingredients** of data visualization, such as colors, line and plot types, etc. (Session 8 / IDS-25)

Instead, we will spend the session on the things that separate an analyst's chart from a published one.

Dos and Don'ts (click links for background)

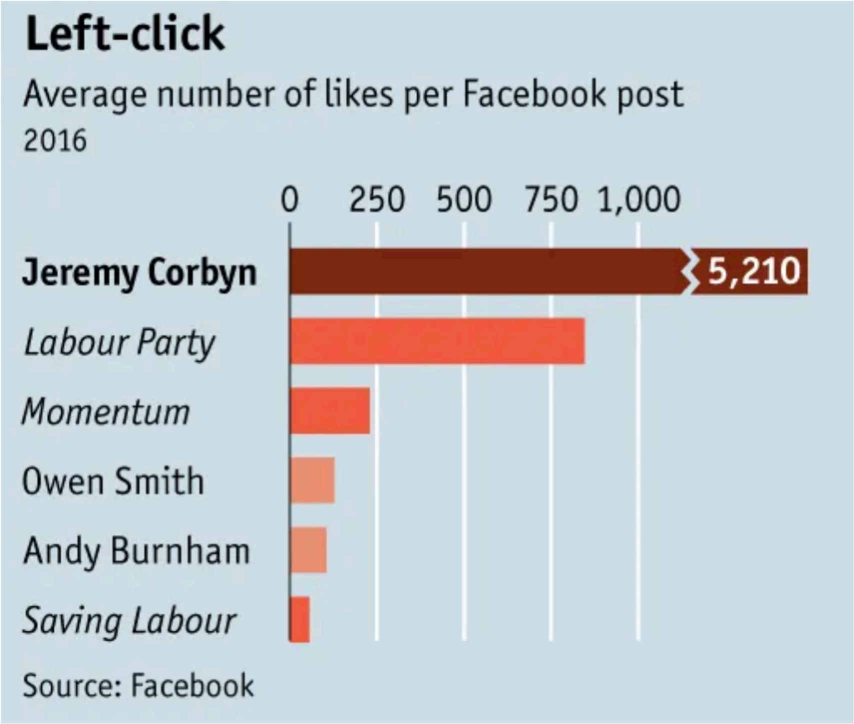
1. Follow the principle of proportional ink.
2. Maximize the data–ink ratio, within reason.
3. Avoid invisible overplotting.
4. Drop all the unimportant stuff.
5. Don't overload graphs. Instead, use several.
6. Use color scales that match the logic of the data scale.
7. Use color-vision deficient-friendly colors.
8. Only use a legend when you need one.
9. Pay attention to legend order.
10. Label axes properly (but avoid trivial information).
11. Use grids and helper lines, within reason.
12. Don't order alphabetically ("Alabama first"). Use natural orders instead.
13. Bar chart axes should include zero.
14. Put the explanatory variable on the x axis, the outcome on the y axis.
15. Axes have canonical directions. Larger values are placed above/right of smaller values.
16. Avoid multiple y axes at all cost.
17. Don't do pie charts. (Or maybe do?)
18. Don't go 3D.
19. Use readable fonts and font sizes.
20. Sometimes a table is just enough.

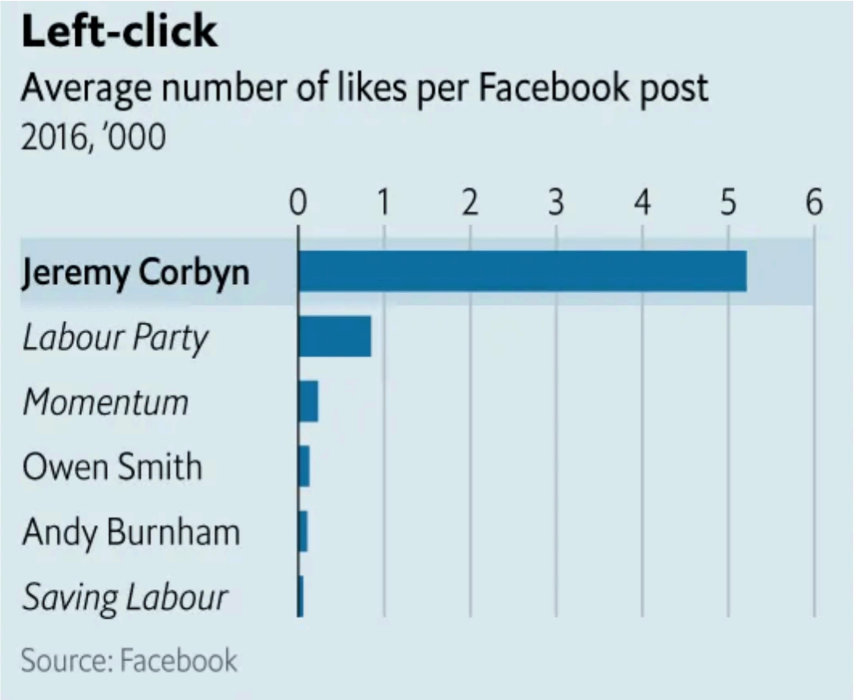
Today

1. Warm-up: two charts, same data
2. Some non-obvious principles
3. Workshop I — fix this chart (15 min, solo, in R)
4. Workshop II — design a chart on the whiteboard (10 min, teams)
5. Your charts: live discussion

Warm-up: two charts, same data

Which do you like better?





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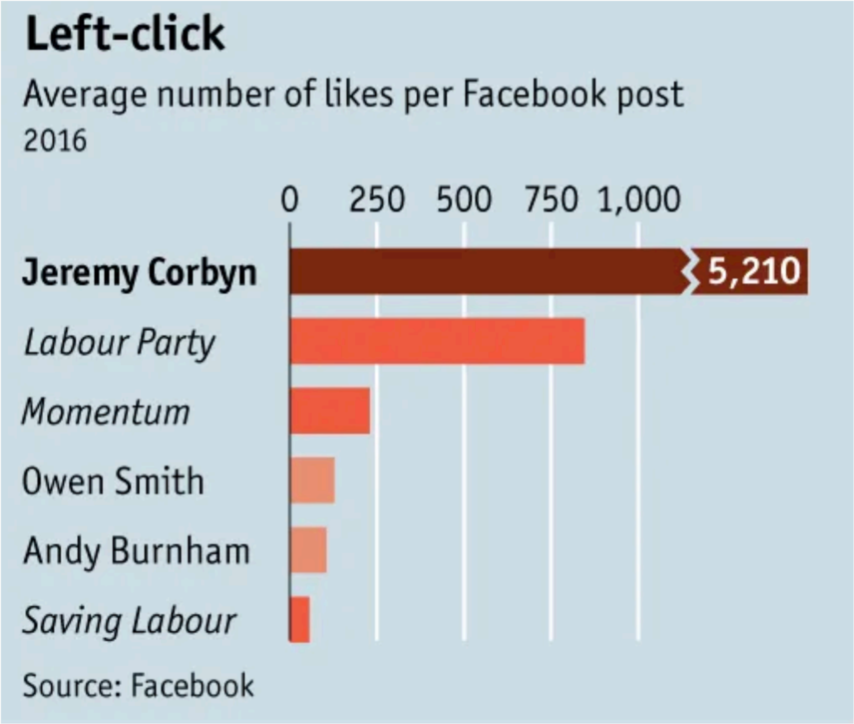
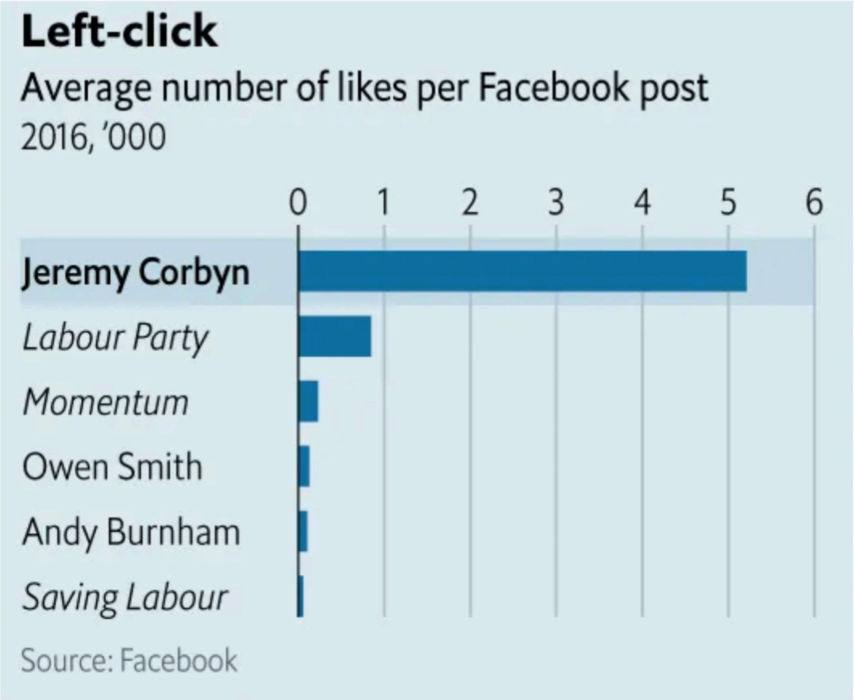
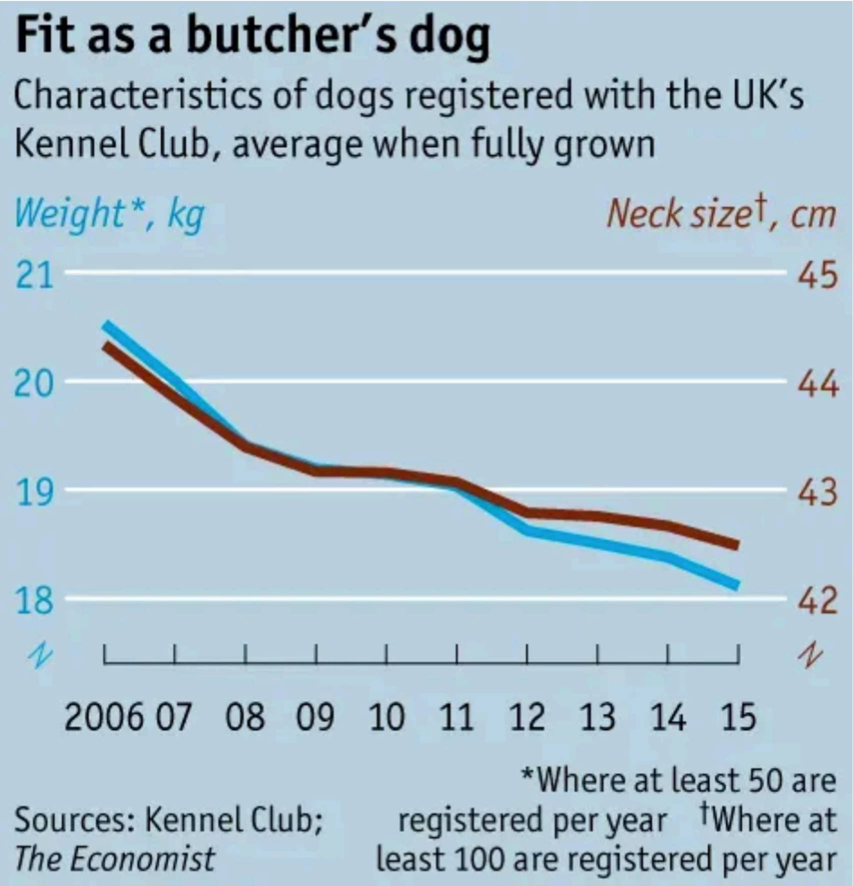
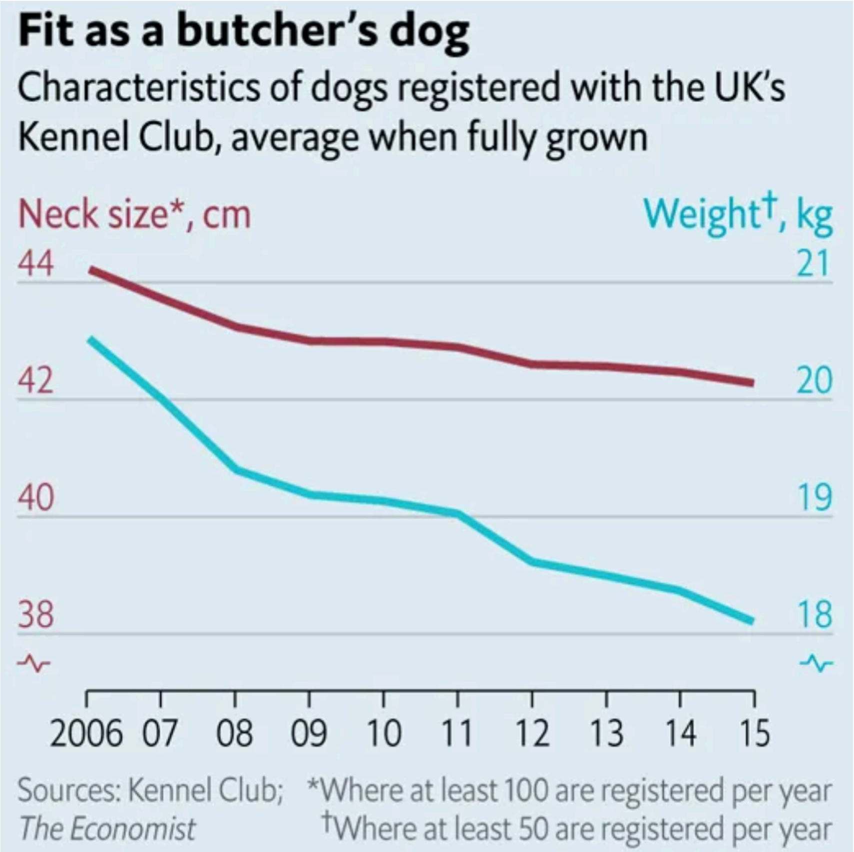




`Source` Sarah Leo, The Economist. Left ❌, right ✅ : truncated scale misleading, colors misleading

Which do you like better?





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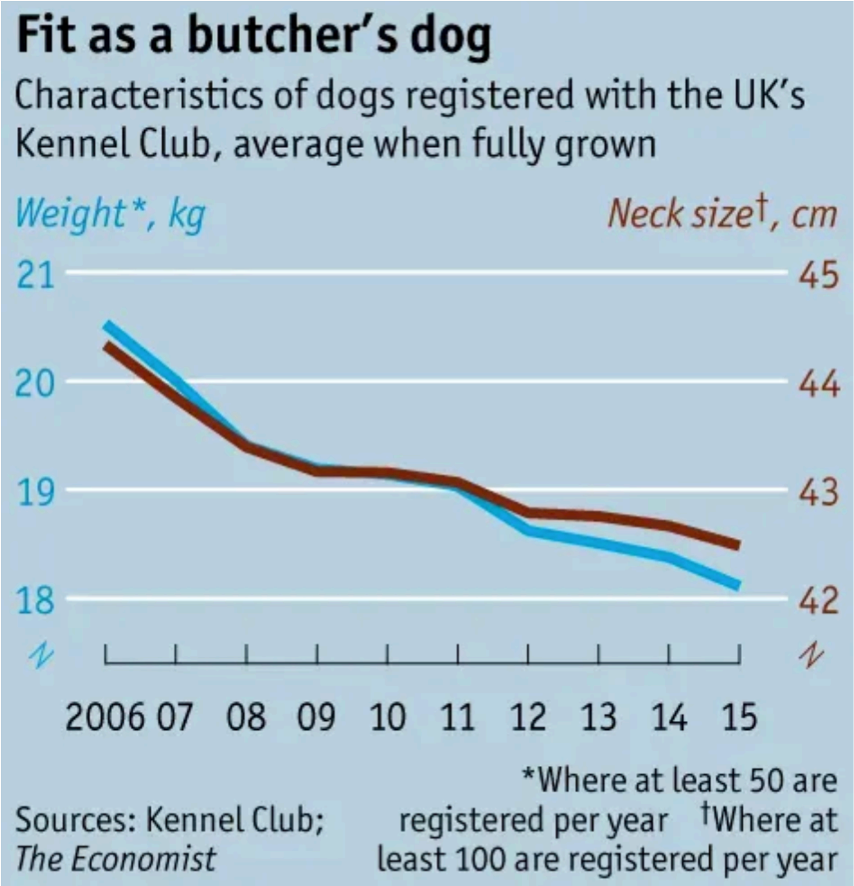
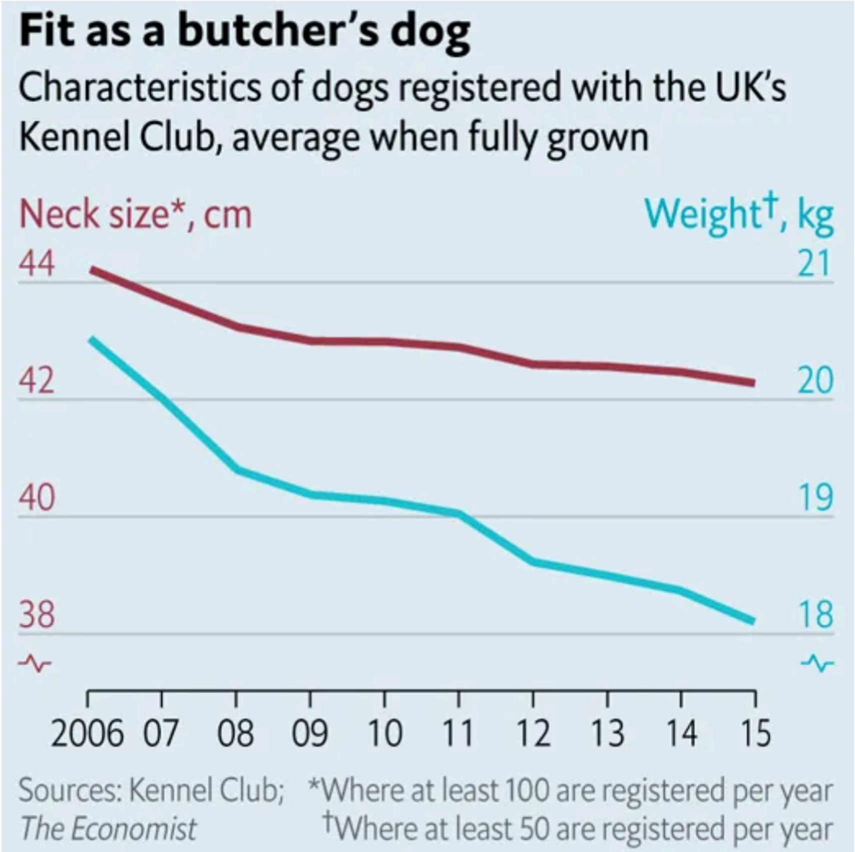
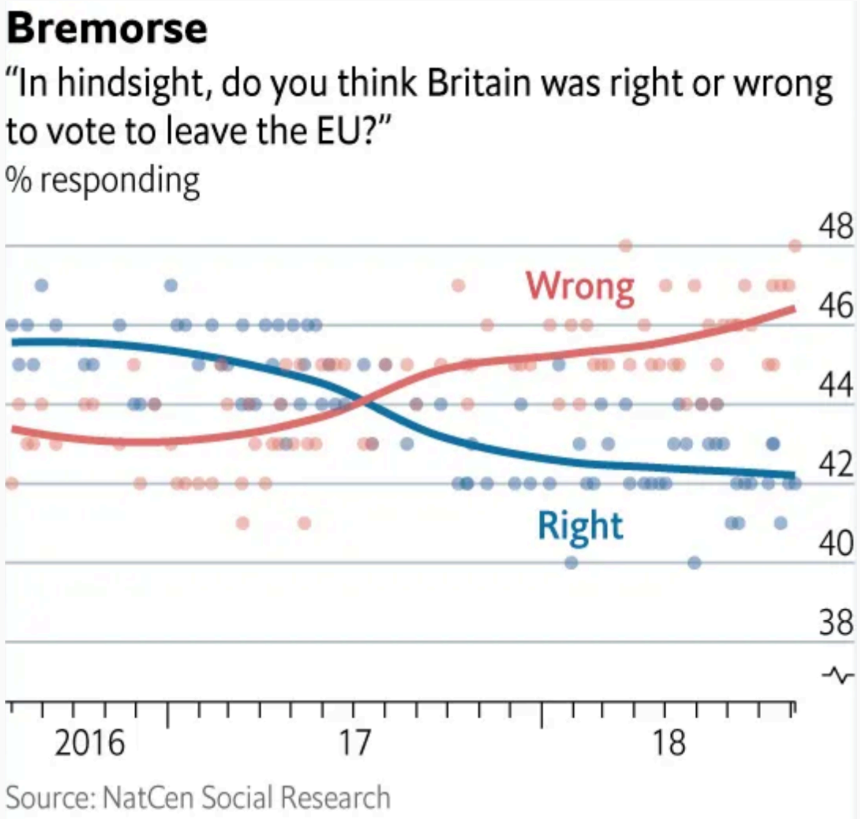
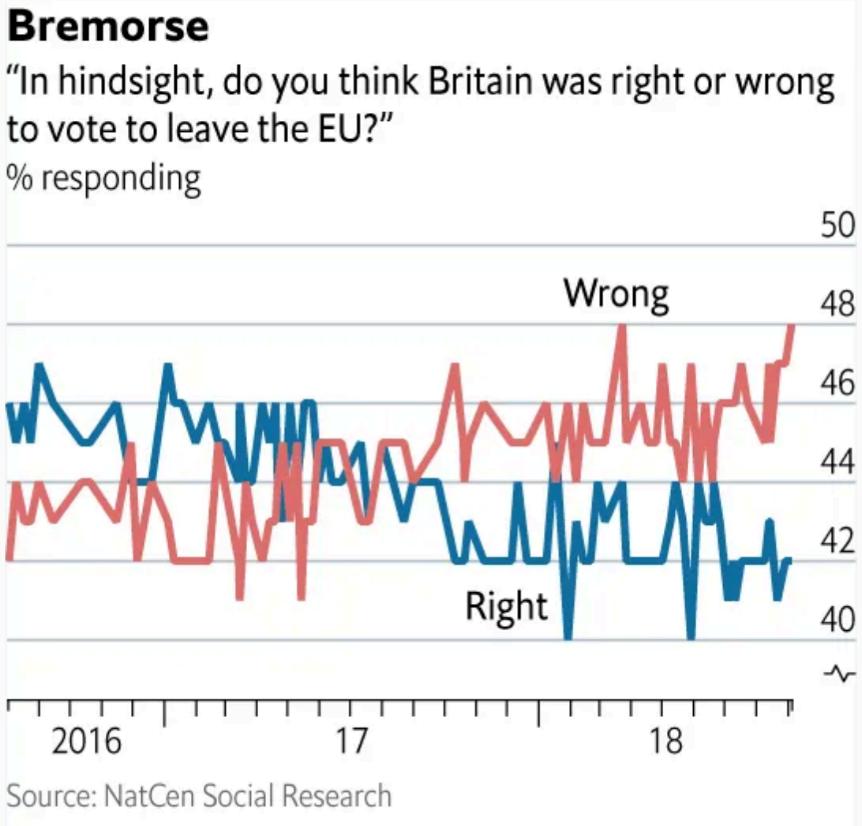




`Source` Sarah Leo, The Economist. Left ❌, right ✅ : scales are cherry-picked, better plot makes proportional change comparable

Which do you like better?





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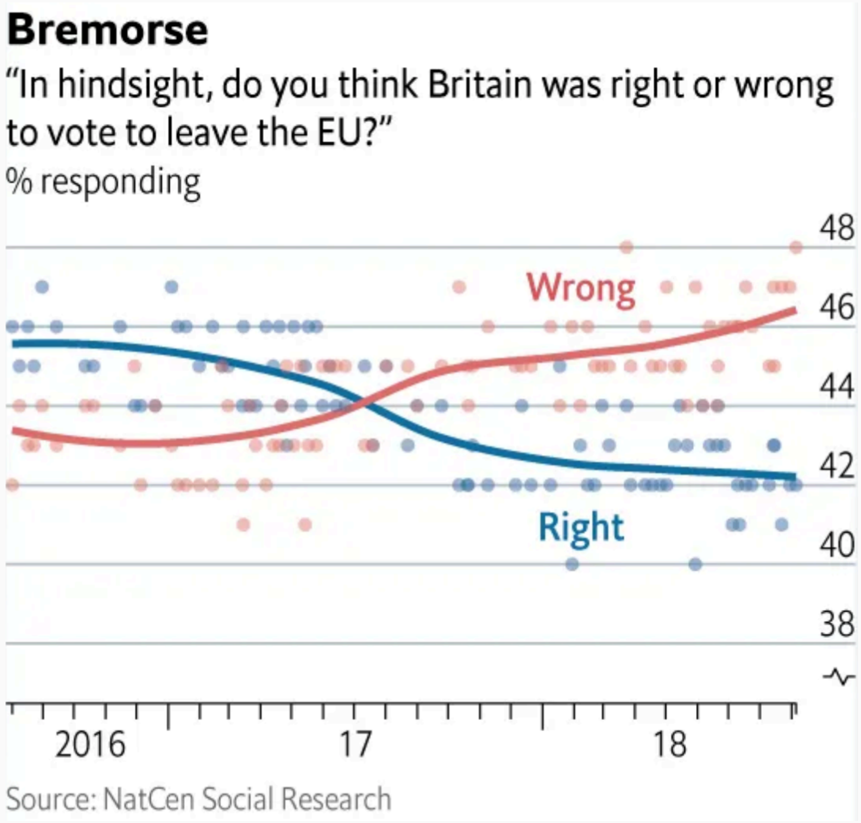
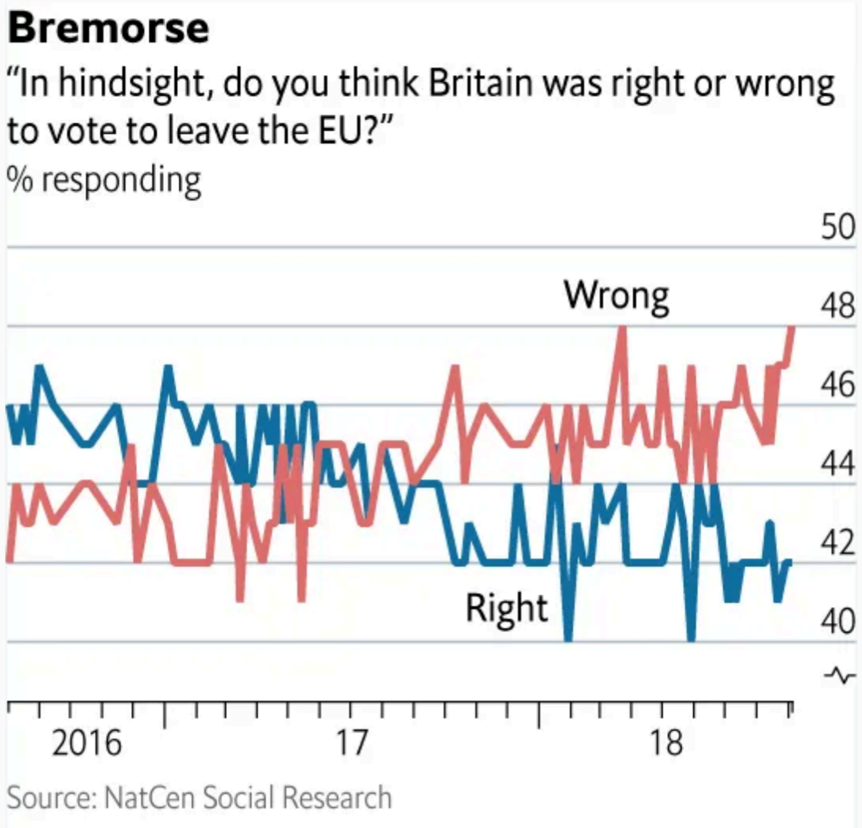
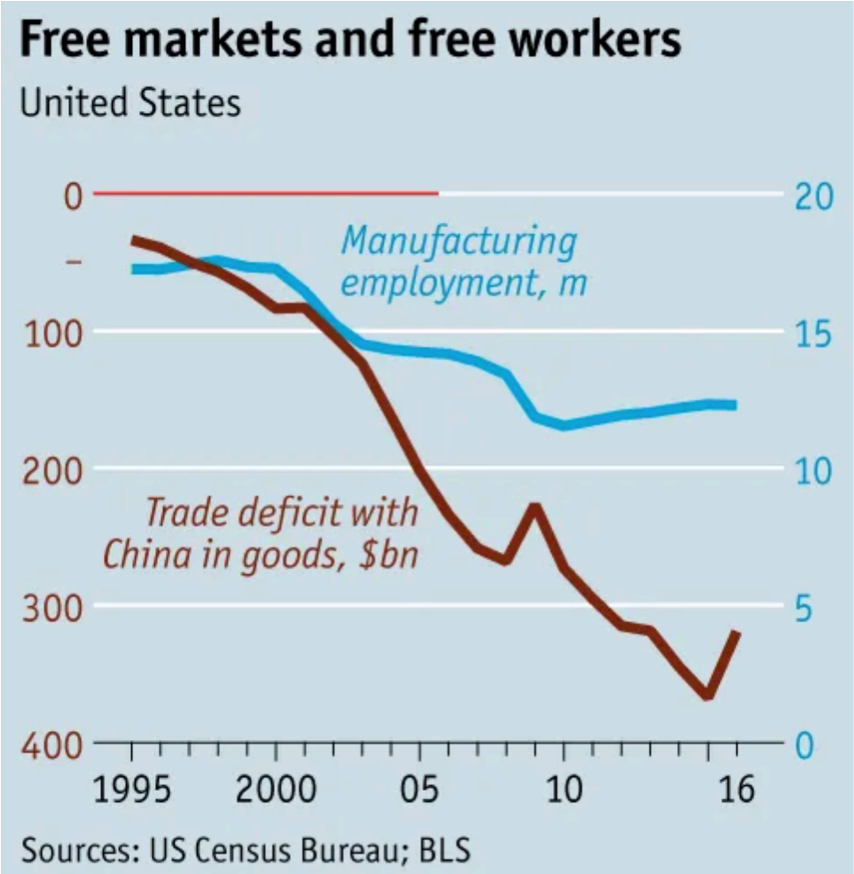
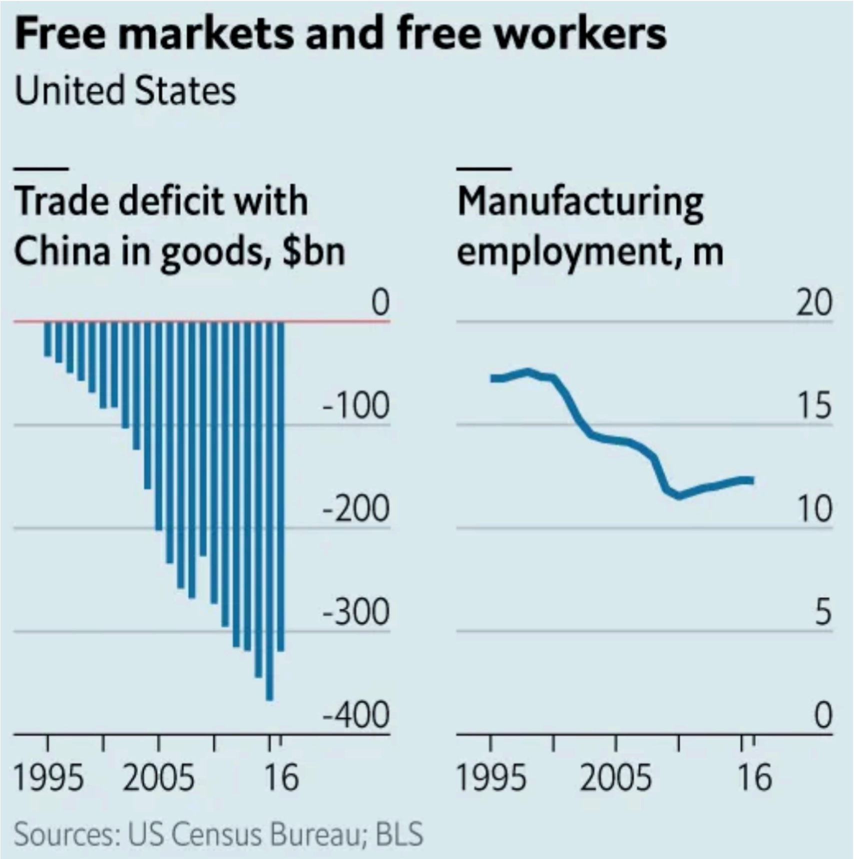




`Source` Sarah Leo, The Economist. Right ❌, left ✅ : poor plot type choice that takes variation in opinion data at face value; also: y scale width.
Francis Gagnon: "The bottom third of the plot area should remain empty for a line chart that doesn't start at zero."

Which do you like better?





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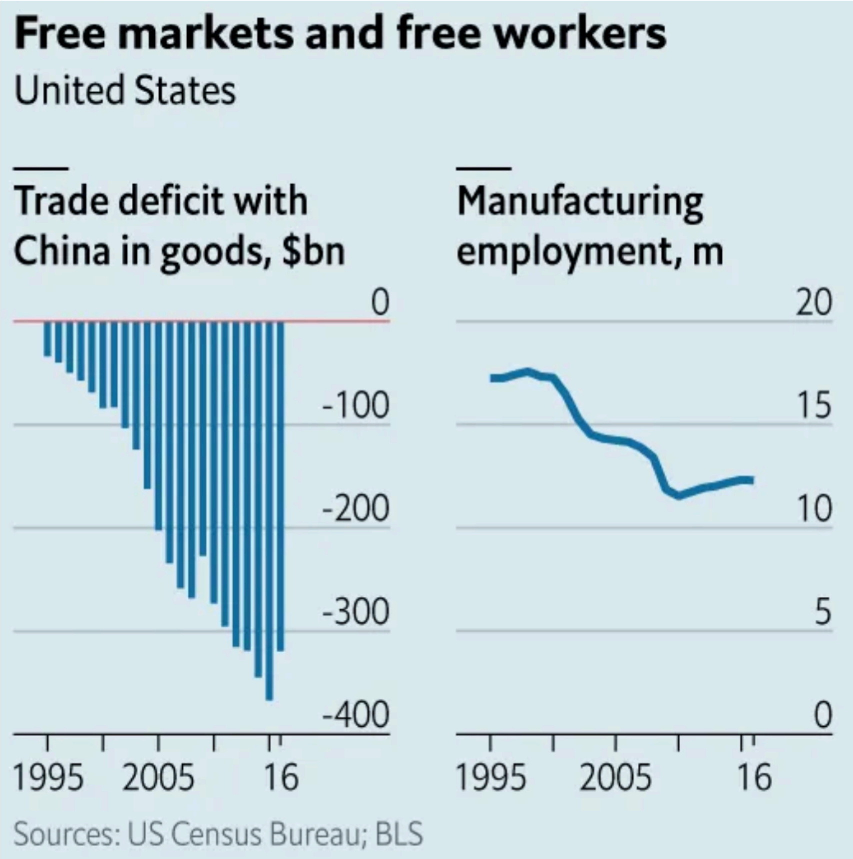

`Source` Sarah Leo, The Economist. Left ❌, right ✅ : difficult to read, combining negative and positive scale

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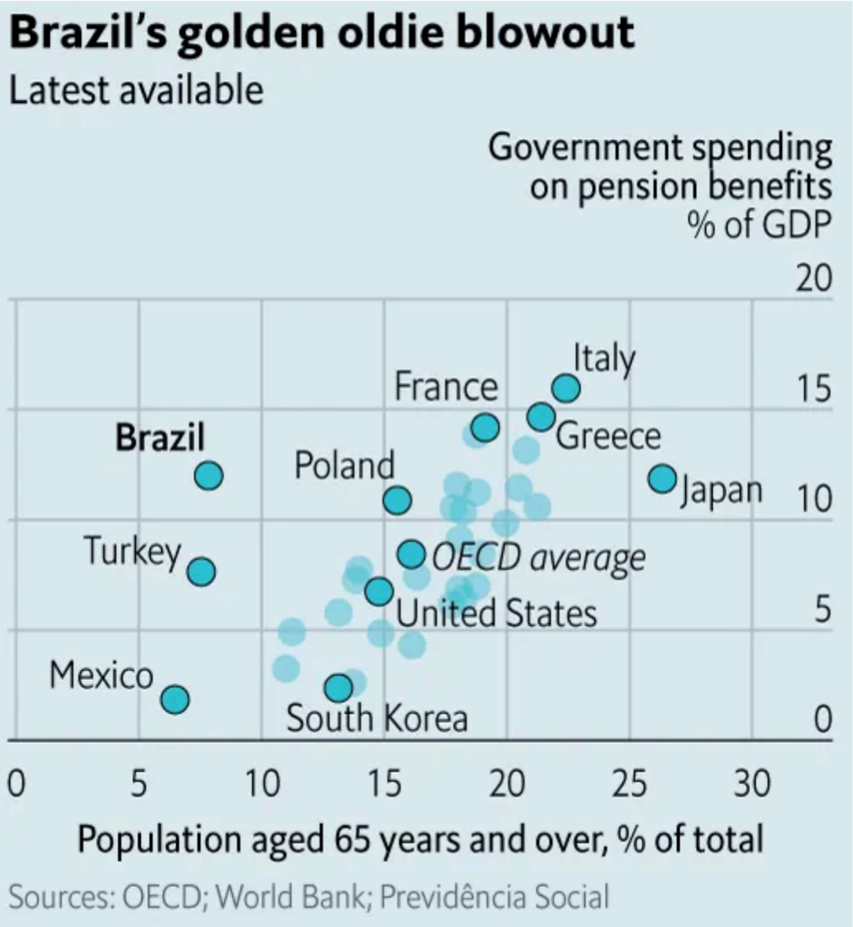

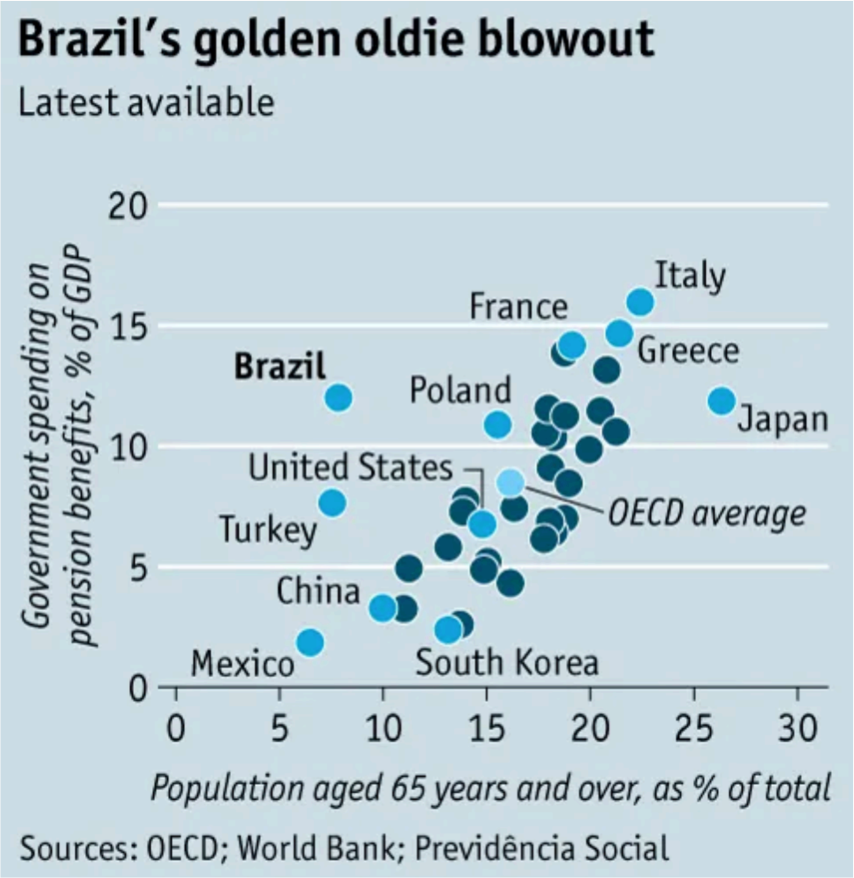

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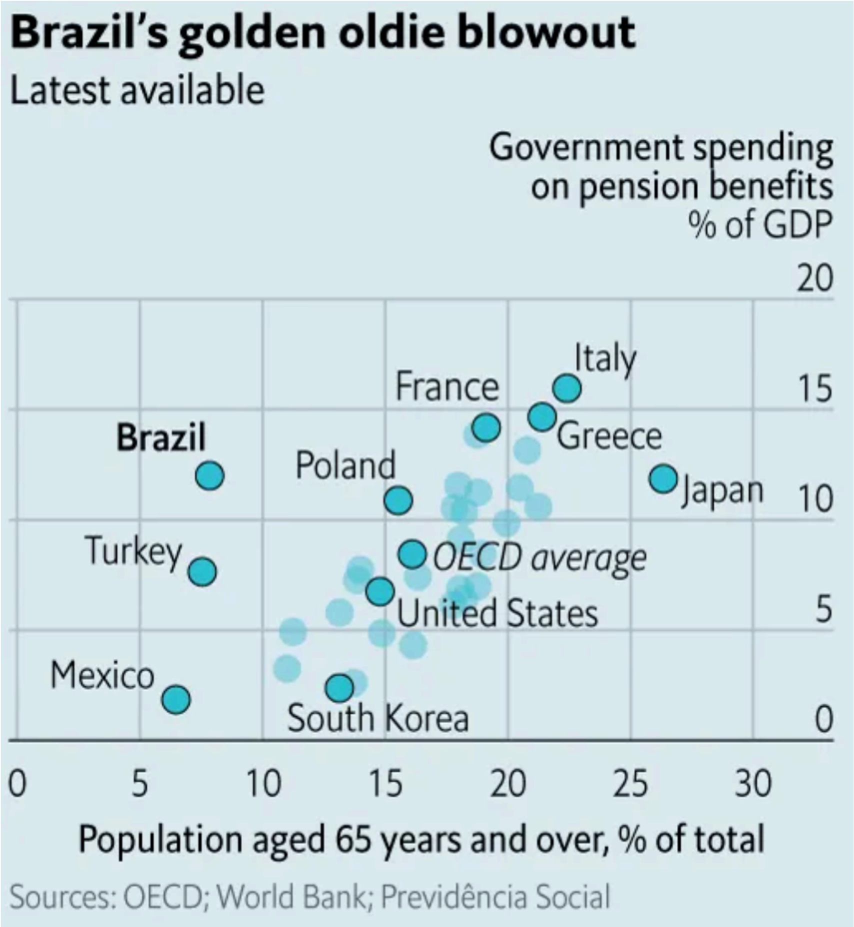

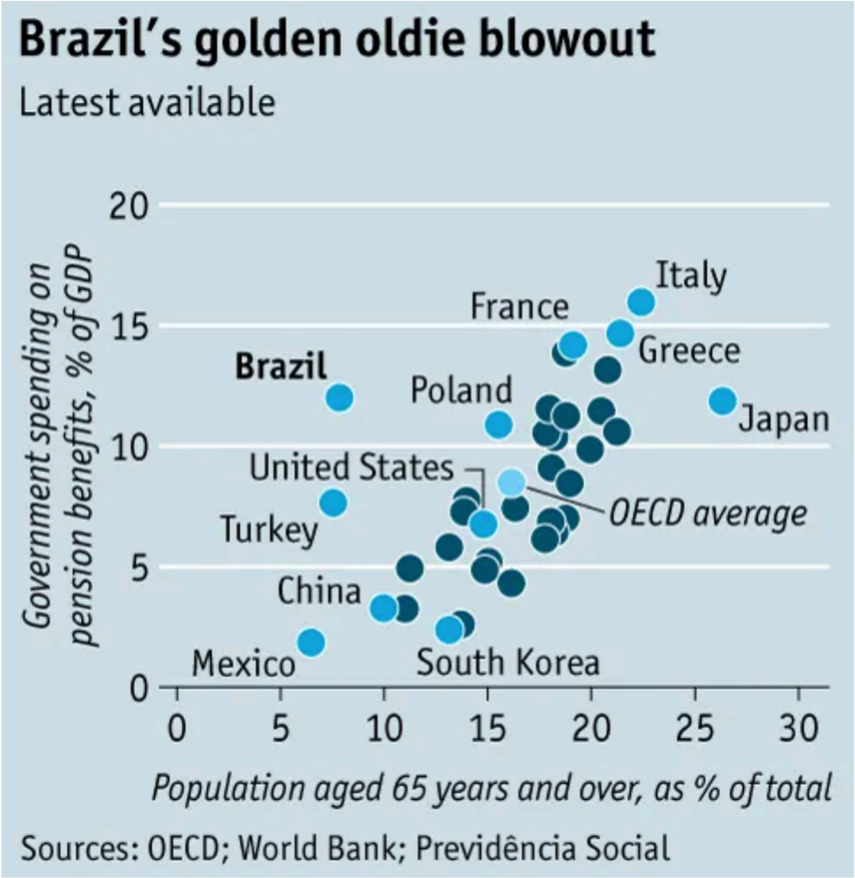

`Source` Sarah Leo, The Economist. Right ❌, left ✅ : colors are not meaningful here;, replaced with opacity; axis title readability; two-way grid

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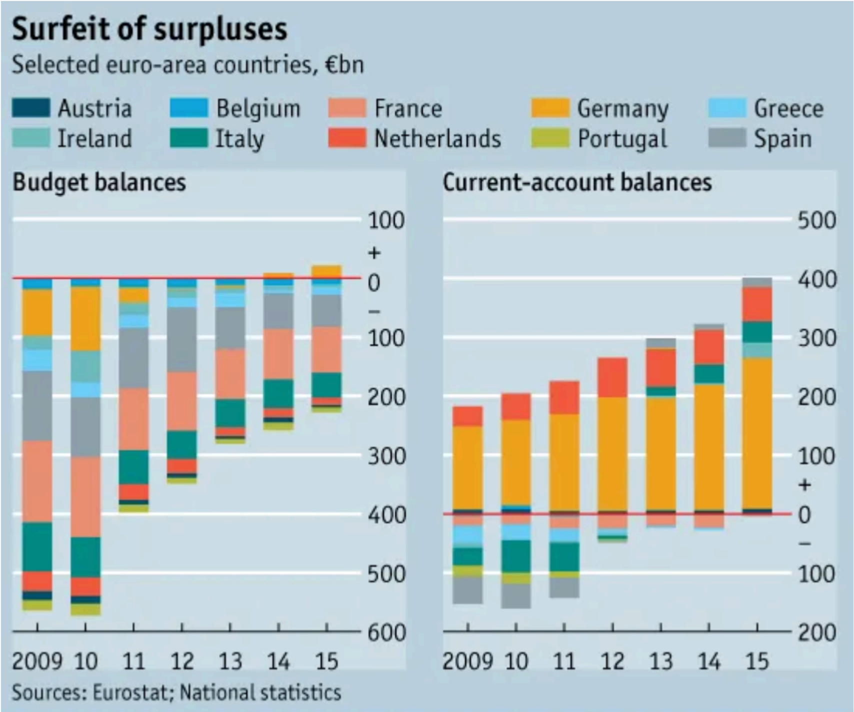

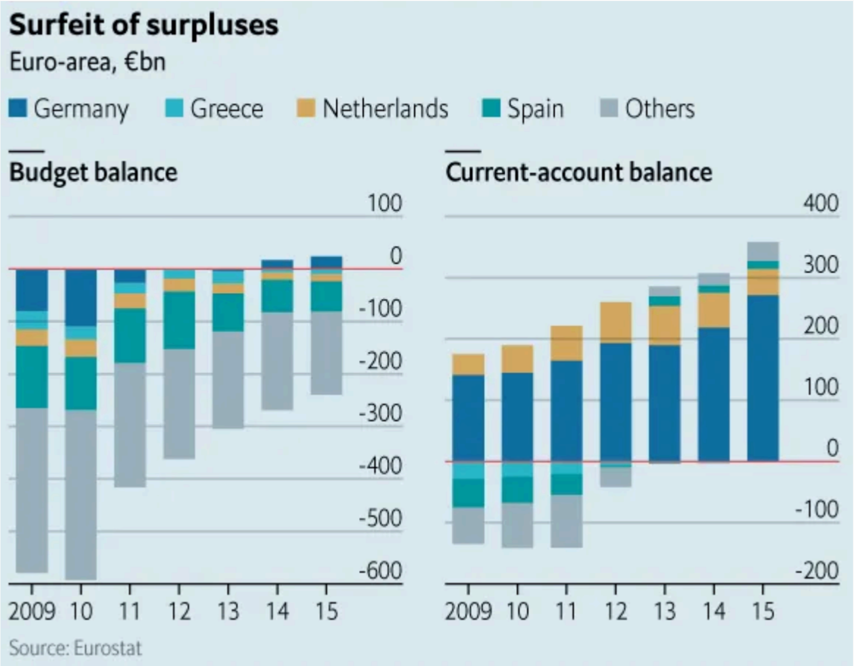

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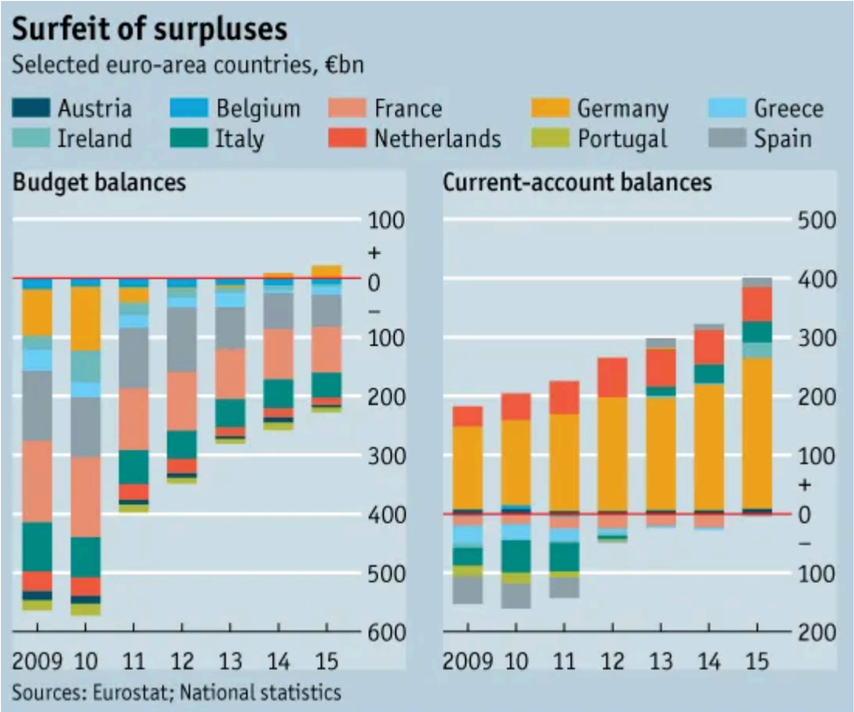
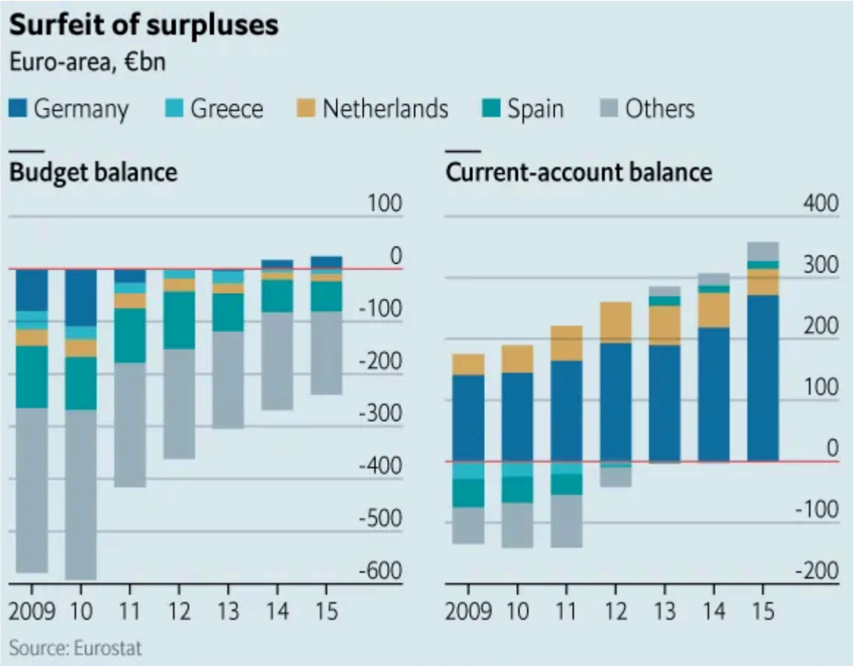


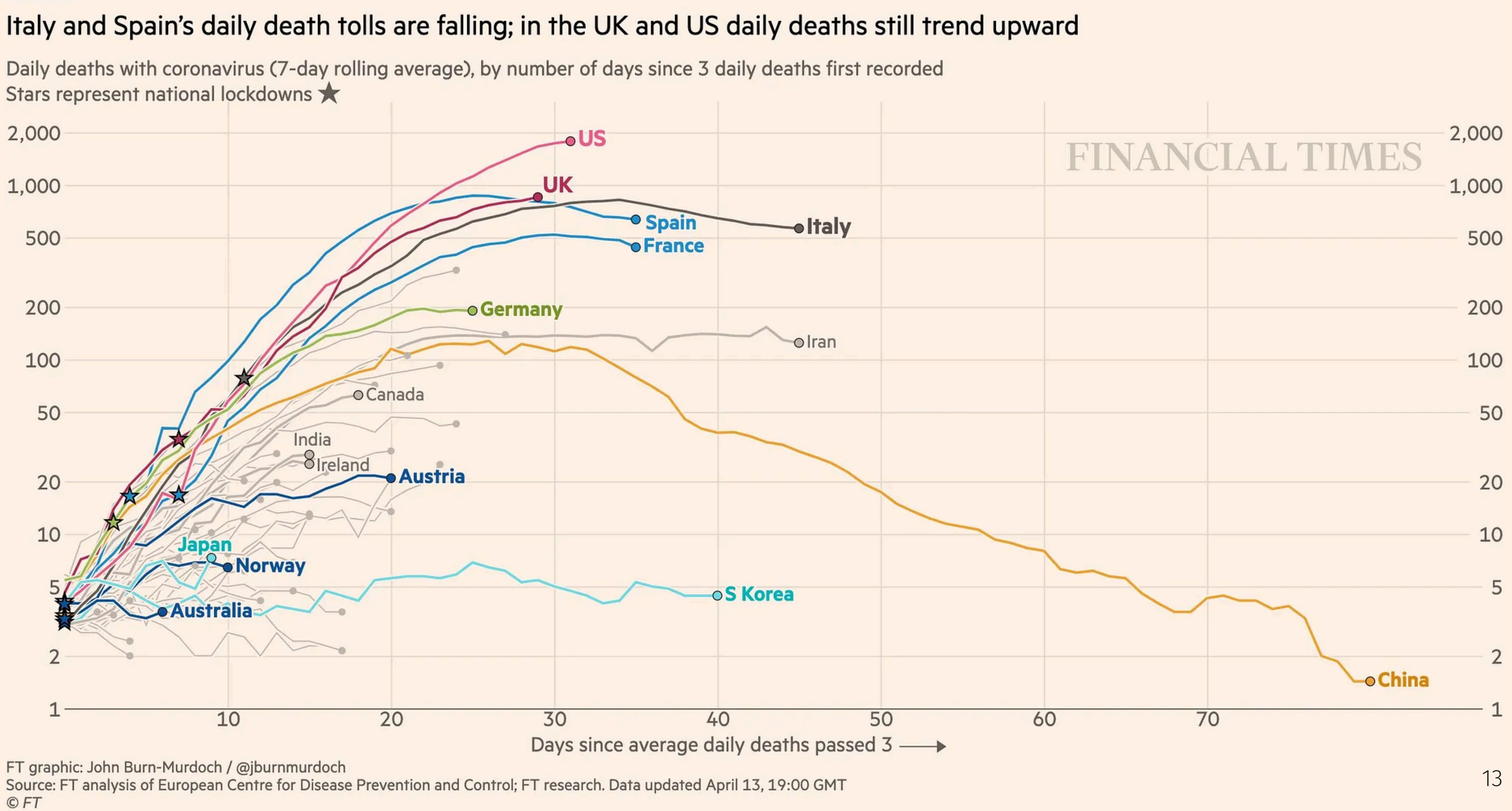


`Source` Sarah Leo, The Economist. Left ❌, right ✅ : too many colors and observations which are not important

Some non-obvious principles

1. Annotation is where the journalism happens

Burn-Murdoch's COVID charts (background here)

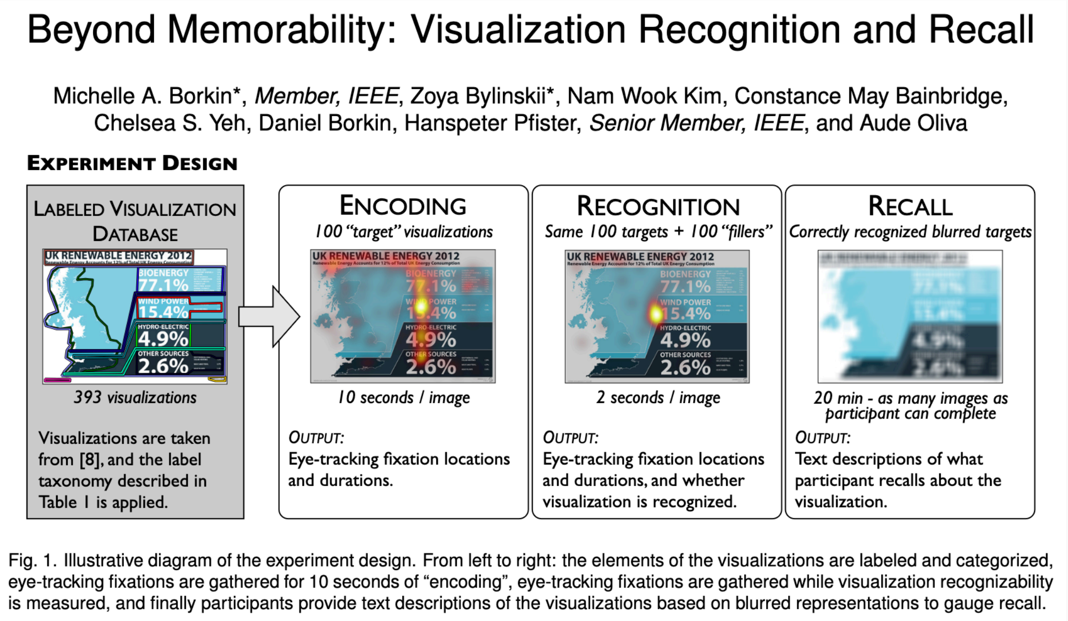


The science of annotations

Borkin et al. (2016) — eye-tracked readers across ~400 charts. What they remembered most was the **title and text**, not the encoding. *IEEE TVCG* · PDF

Kong, Liu & Karahalios (2018) — same chart, different titles. A title that slants toward one interpretation **shifts what readers take away**, even when the data is identical. A misaligned title can override the perception entirely. *ACM CHI* · PDF

Hullman & Diakopoulos (2011) — frames annotation as one of **four editorial layers** (data, representation, annotation, interactivity). The annotation layer is *where rhetoric lives*, and where it is most often denied. *IEEE TVCG* · PDF



`Source` Borkin et al. (2016)

> The words on a chart aren't decoration — they *are* the message that is most likely to be remembered.

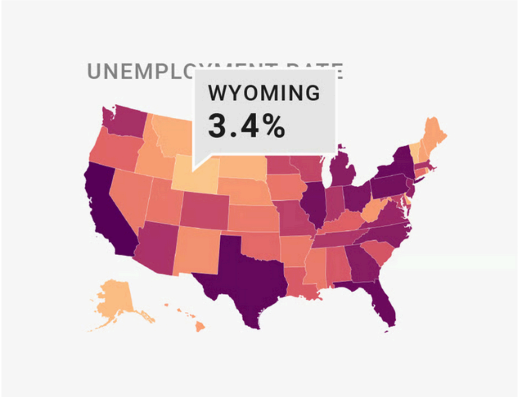
What to consider when using text (Lisa Muth, Datawrapper)

Label directly

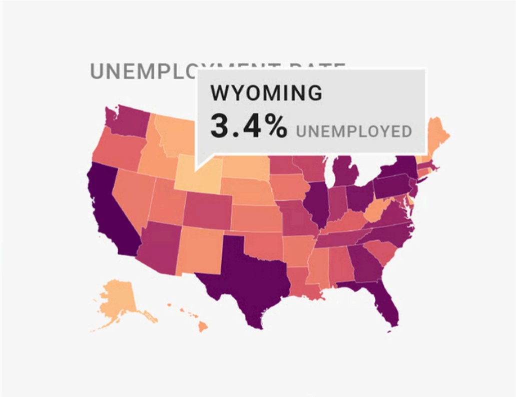
NOT IDEAL

BETTER

Remind people what they're looking at in tooltips

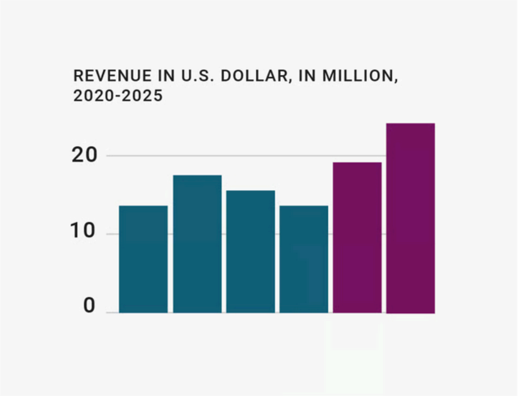


NOT IDEAL



BETTER

Repeat the units your data is measured in



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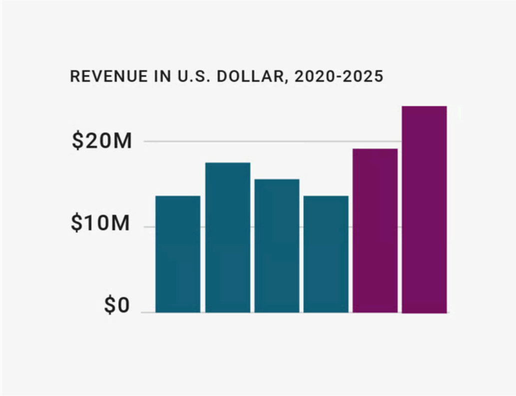
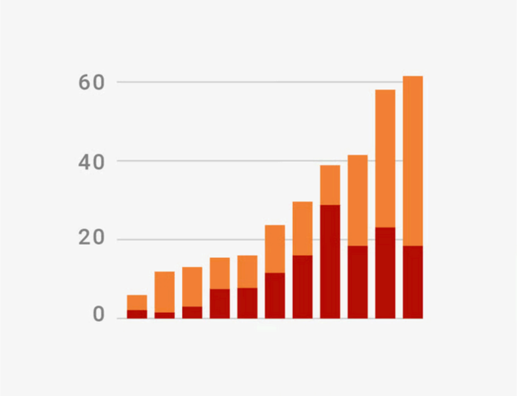


BETTER

Move the axis ticks where they're needed



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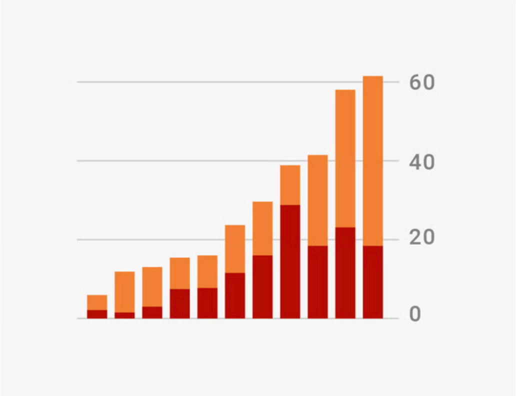
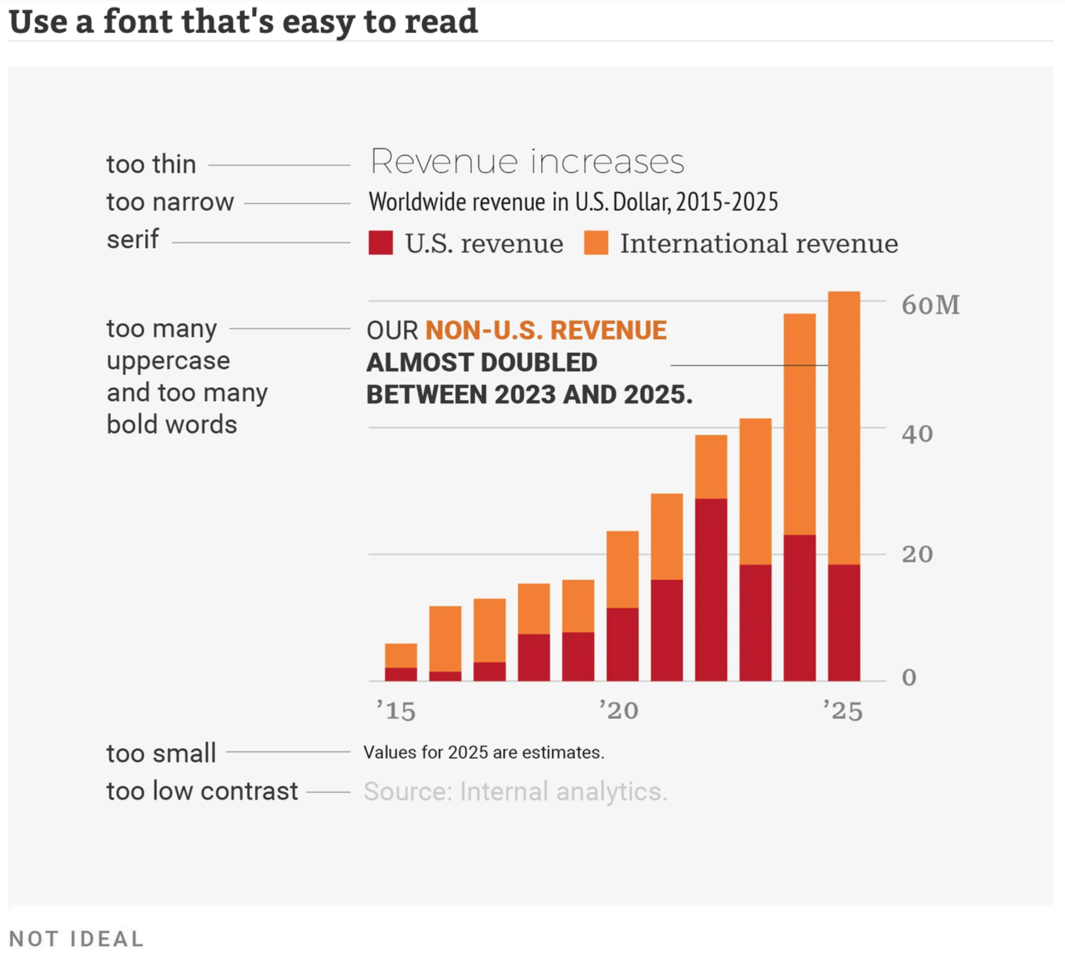
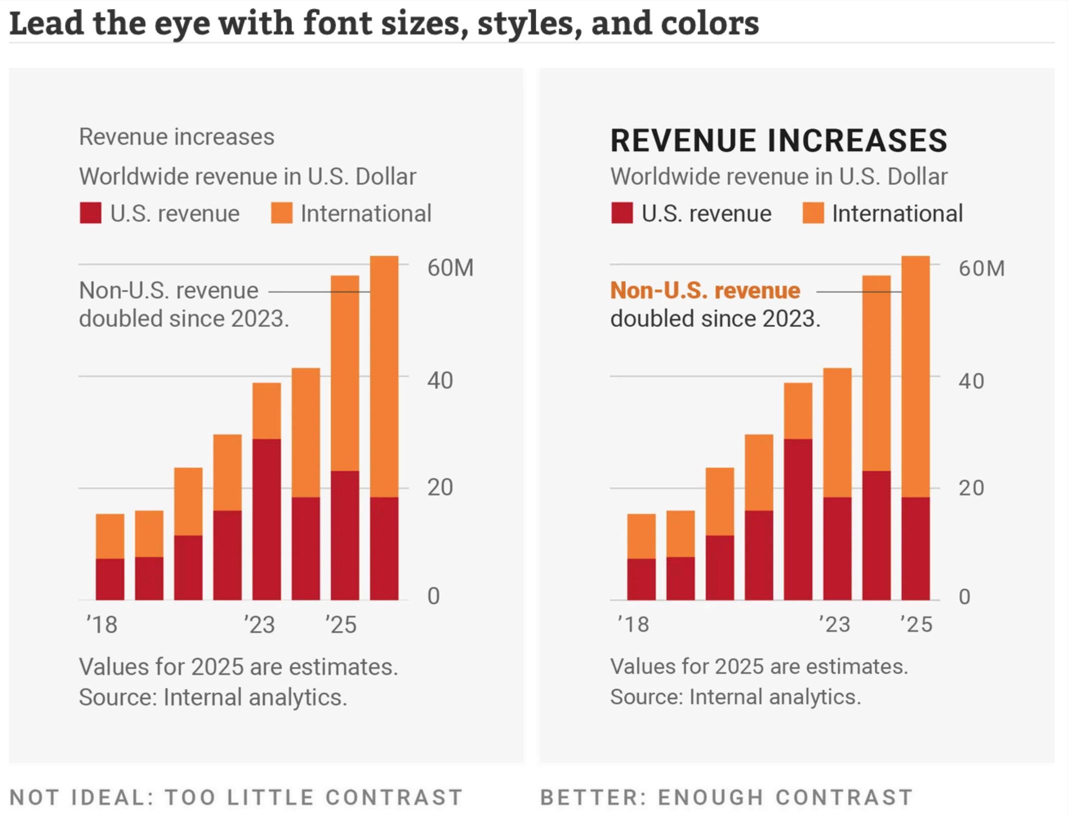


BETTER

What to consider when using text (Lisa Muth, Datawrapper)



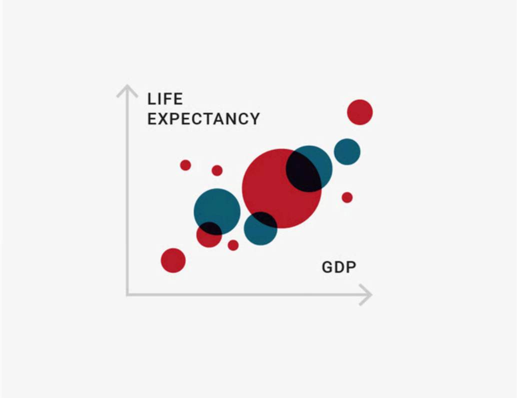


More on which fonts to use in Datawrapper's typography guide here (great read with examples!).

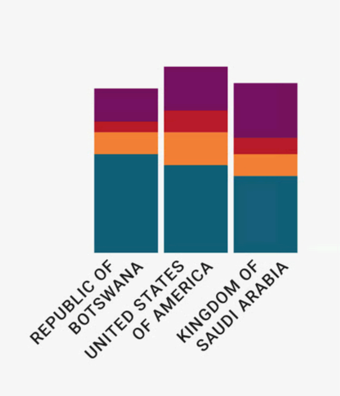
What to consider when using text (Lisa Muth, Datawrapper)

Don't make your readers turn their heads

NOT IDEAL



BETTER



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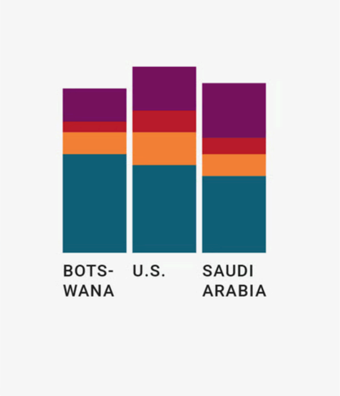
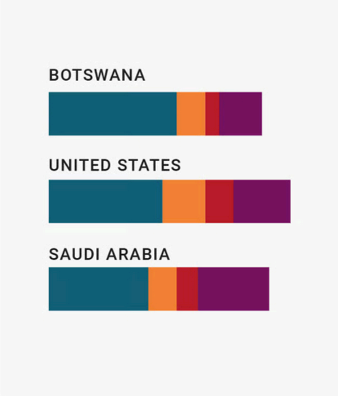
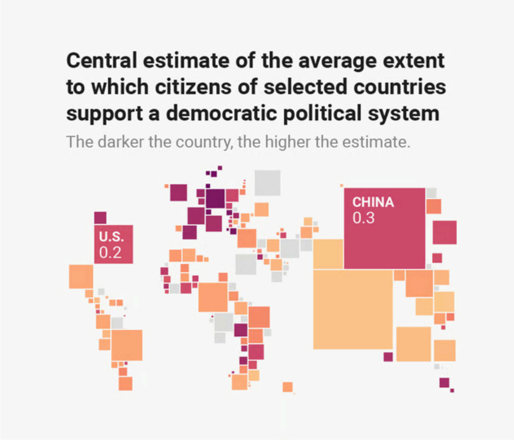


BETTER



EVEN BETTER

Use straightforward phrasings



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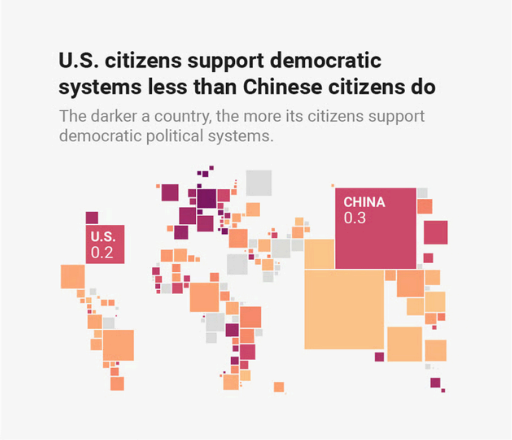
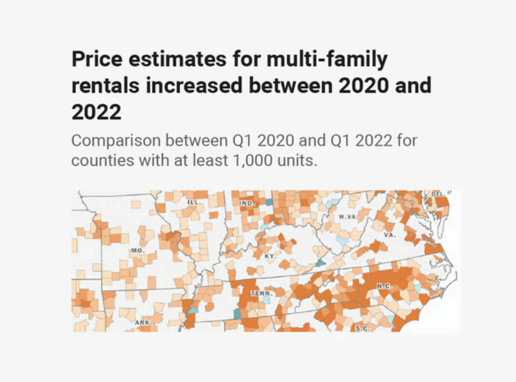


BETTER

Be conversational first and precise later



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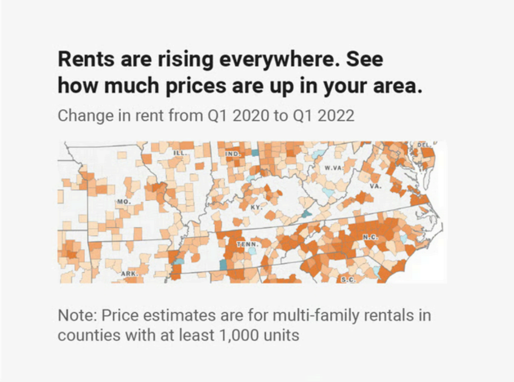
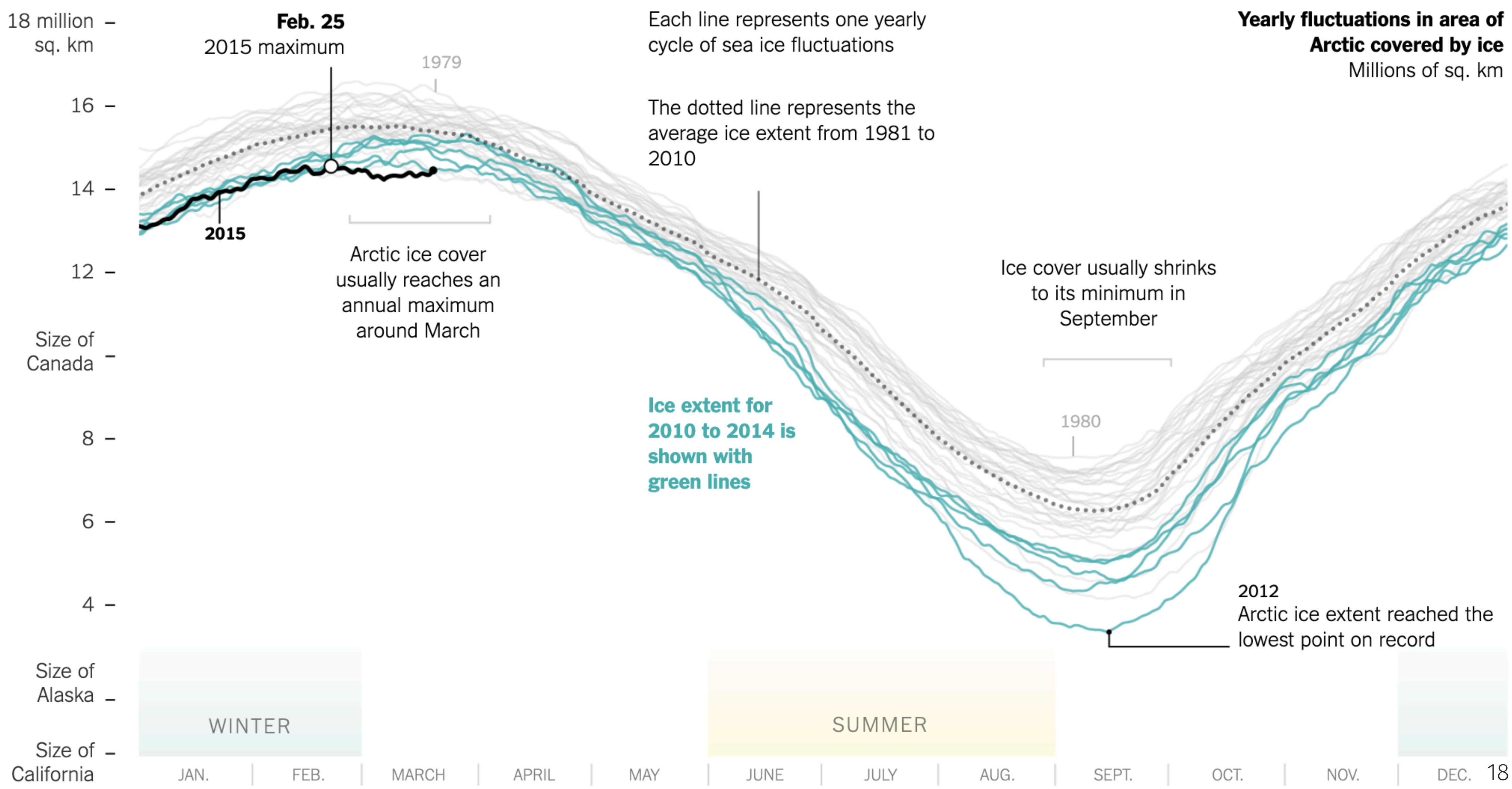


BETTER

Annotation up to 11 (Derek Watkins for NYT, 2015)



Workshop I (15 min, solo)

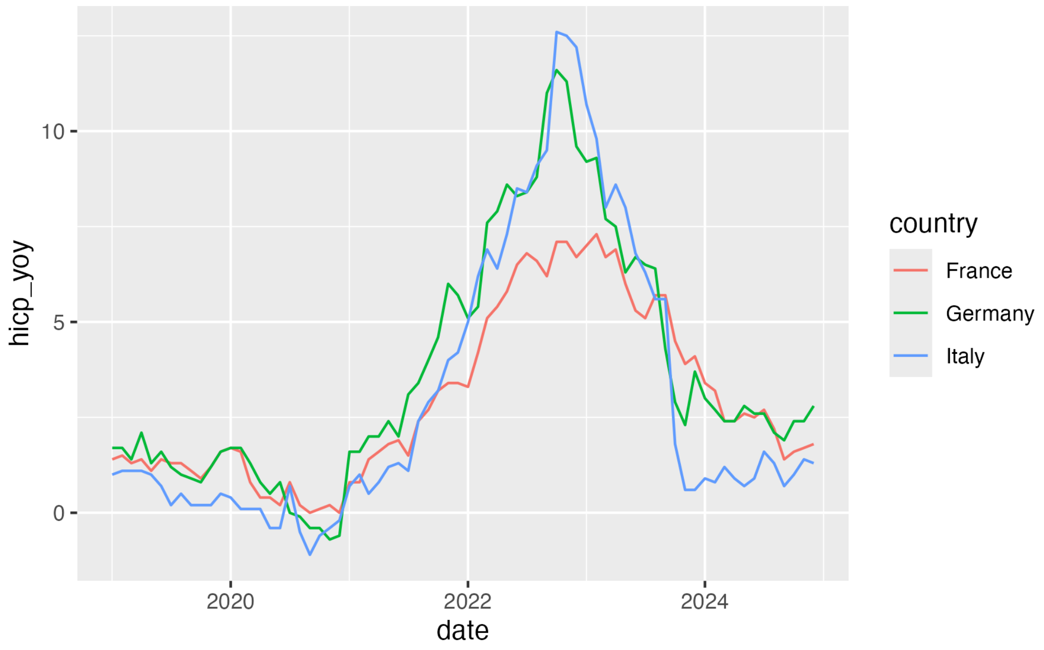
Workshop I — fix this chart

Materials

In `code/visualization/` (and on Moodle) you'll find:

- a small dataset on **monthly year-on-year inflation, 2019–2025** (Eurostat HICP),
- a script `workshop-improve-this-chart.R` that creates a deliberately bad ggplot chart,
- a `# Your turn` block.

Figure 1



Task (15 min, on your own)

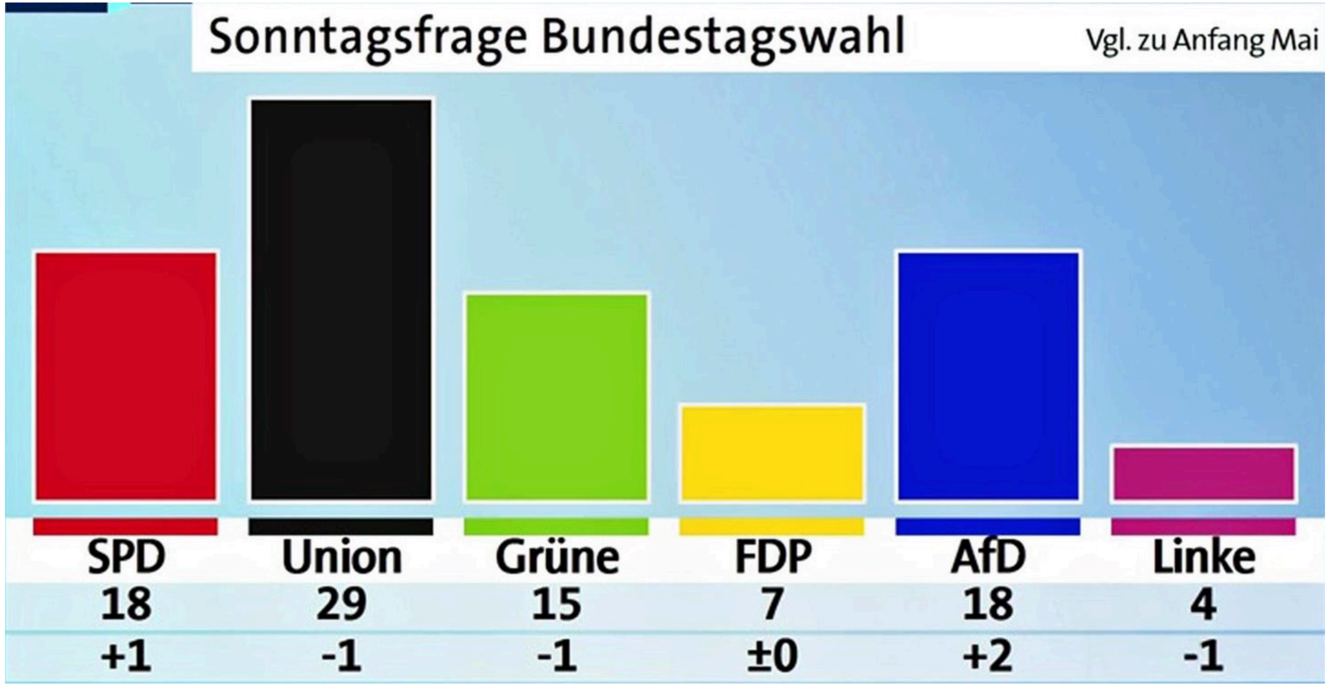
Turn the analyst's chart into a journalist's chart. Aim for *one* clear headline finding.

We'll come back together at the 15-minute mark.

2. Defaults are editorial choices

Guess the outlet!

Score: 0 / 0



ARD | Bild | The Economist | FiveThirtyEight | Financial Times | The Guardian | Our World in Data

Pew Research Center | Süddeutsche Zeitung | ZDF | Die Zeit

House styles

What is a house style?

A house style is *internally consistent* and *externally recognisable*. Both matter.

Tools are not neutral

Every tool encodes assumptions about what is *normal*:

- `ggplot2`'s grey panel and serif title: "this is a plot for analysts".
- Excel's pastel gradient: corporate-meeting-coded.
- Datawrapper's blue–orange diverging: newsroom-neutral, but it *is* a choice.

Each of these encodes who the chart is *for*. Choosing them passively is choosing — you just didn't know you were choosing.

What this means for you

You don't have a house style — *yet*. But every chart you publish is training your audience to expect one. So:

1. Pick a small, fixed palette (3–5 colours, plus grey for "context" data) and stick to it across pieces.
2. Pick *one* font and use it everywhere. Default sans is fine.
3. Decide once: do your titles end in a period? Do your axis labels include units? Where does the source line live?
4. Write a `theme_yourname()` and reuse it.

Don't reinvent the wheel — start from a base theme

Built into ggplot2

`theme_grey()` *(default)*, `theme_bw()`, `theme_minimal()`, `theme_classic()`, `theme_void()`, `theme_dark()`, `theme_light()`.

```
R> ggplot(mpg, aes(displ, hwy, colour = class)) +
+   geom_point() +
+   theme_minimal()
```

Pre-baked publication looks: `ggthemes`

```
R> install.packages("ggthemes")
R> library(ggthemes)
```

You get `theme_economist()`, `theme_fivethirtyeight()`, `theme_wsj()`, `theme_stata()`, `theme_excel_new()`, `theme_tufte()`, `theme_few()`, ... — each modelled on a real outlet's house style. I recommend none of them.

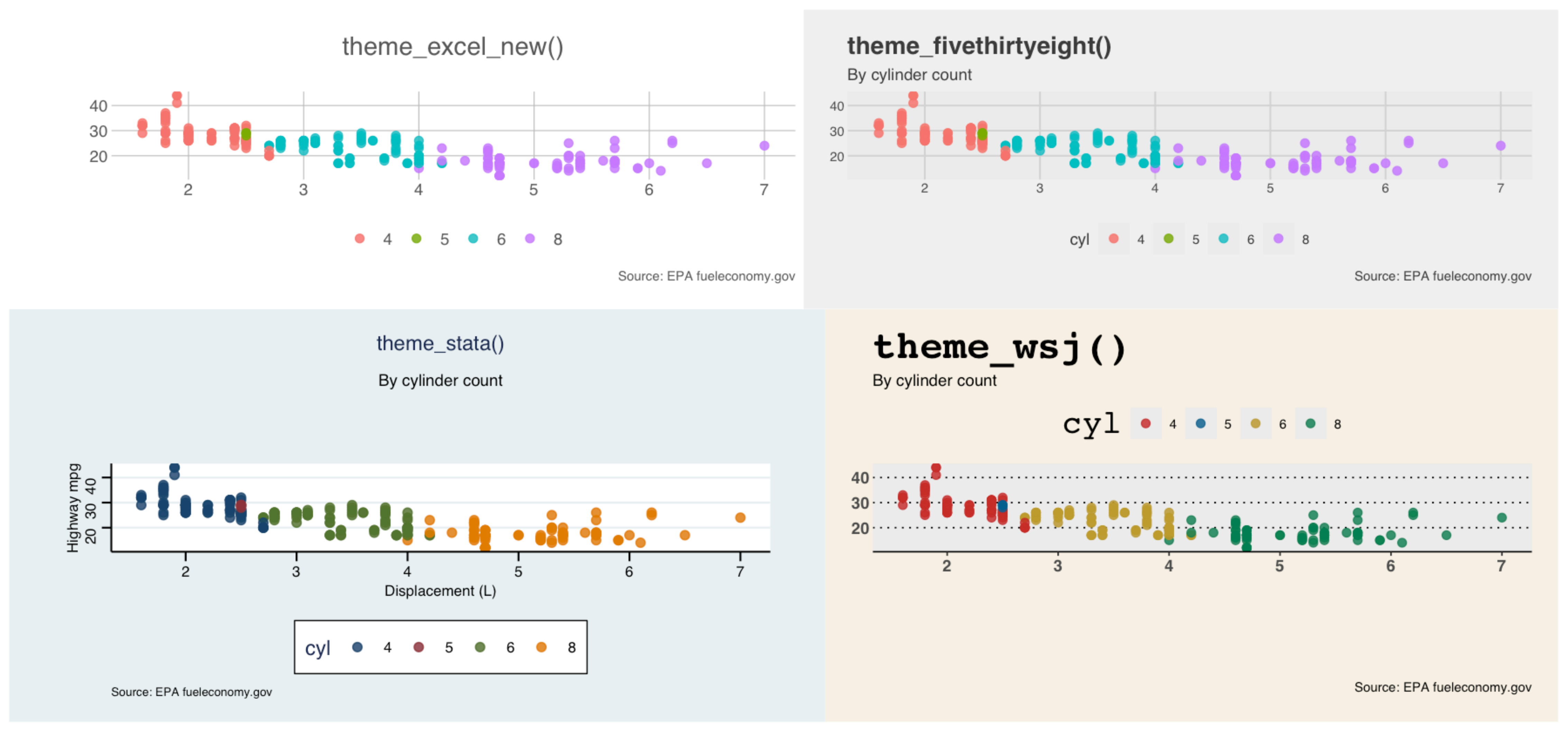
Why bother with a base?

A base theme decides **80%** of the boring stuff for you — panel background, axis lines, gridlines, default font sizes — so you can focus on the **20%** that's editorial: palette, type, annotation placement.

A common workflow

1. Pick a base (`theme_minimal()` is a good neutral starting point).
2. Override the few elements that matter for *your* style (`theme( ... )`).
3. Save the combination as `theme_yourname()`.
4. Re-use across every chart in a project.

Same plot, four `ggthemes`



Roll your own `theme_yourname()`

A custom theme is just a function that returns a `theme()` object on top of a base.

```
R> theme_dj <- function(base_size = 12) {
+   theme_minimal(base_size = base_size) +
+     theme(
+       plot.title.position   = "plot",
+       plot.title            = element_text(
+         face = "bold", size = rel(1.15),
+         margin = margin(b = 4)),
+       plot.subtitle         = element_text(
+         colour = "grey30",
+         margin = margin(b = 14)),
+       plot.caption.position = "plot",
+       plot.caption          = element_text(
+         colour = "grey45", hjust = 0,
+         size = rel(0.7),
+         margin = margin(t = 12)),
+       panel.grid.minor      = element_blank(),
+       panel.grid.major.x    = element_blank(),
+       axis.text             = element_text(colour = "gr
+     )
+ }
```

Your theme in action

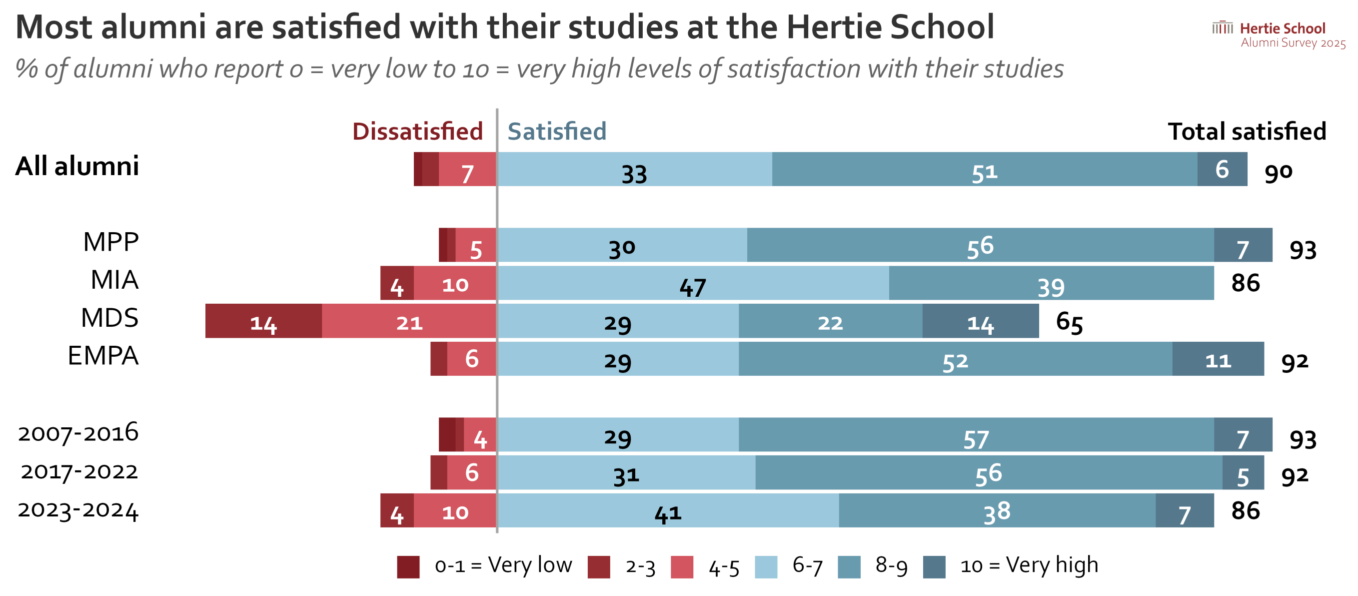
```
R> ggplot(mpg, aes(displ, hwy)) +
+    geom_point() +
+    labs(title    = "Bigger engines, lower mileage",
+         subtitle = "EPA fuel-economy data, 1999–2008",
+         caption  = "Source: EPA fueleconomy.gov") +
+    theme_dj()
```

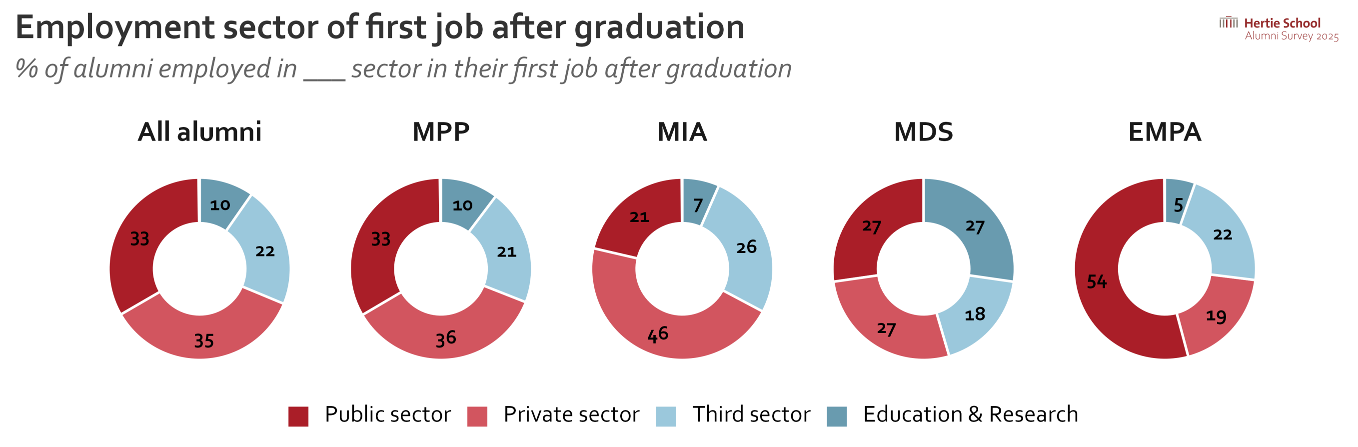
Two patterns worth stealing

- Use `rel()` for sizes so everything scales when you change `base_size`.
- Set `plot.title.position = "plot"` so the title and caption align with the plot edges, not the panel — a small detail readers feel without naming.

Once your theme is stable, drop it in a tiny `R/theme_dj.R` and `source()` it from every report. Same `theme_dj()`, every chart.

Example: Hertie brand for alumni survey



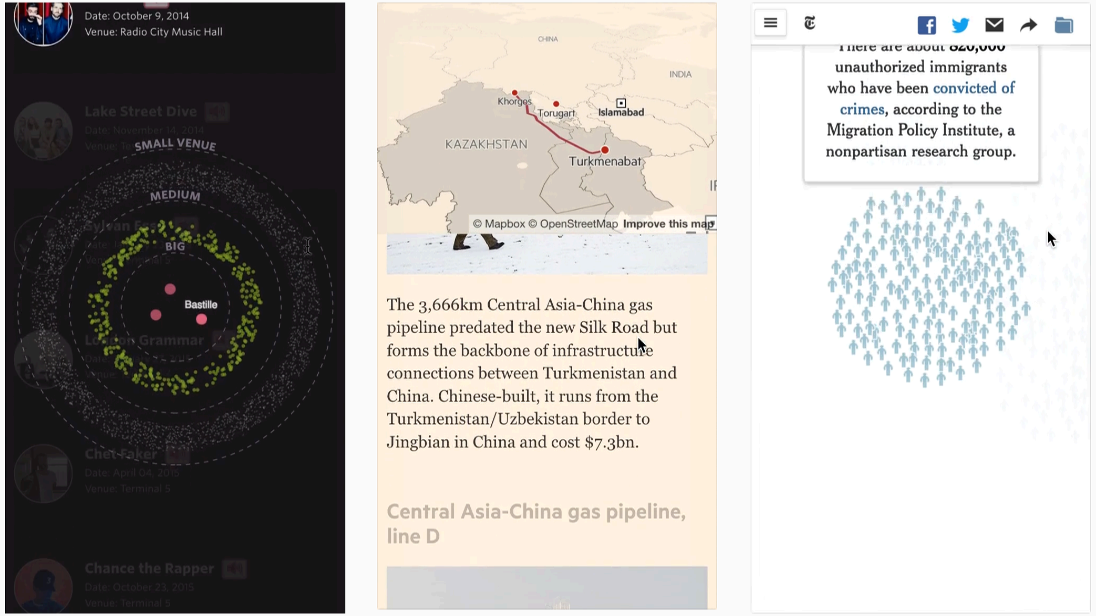


3. Interactivity has to earn its place

Most readers do not interact

What the research and newsroom analytics find

- **Boy, Détienne & Fekete (CHI 2015)** ran three field experiments on a major news site and a viz gallery: introductory "story" framing on an interactive chart **did not** increase subsequent exploration. (HAL preprint · ACM)
- **Archie Tse, NYT (Malofiej 2016)** — "Why we are doing fewer interactives". Internal NYT data: most clickable elements got <10% engagement. (Nieman Lab summary · slides)
- **Gregor Aisch, NYT (2017)** — pushes back, but confirms the underlying number: **~85% of readers** of NYT graphics from 2015 did not click any button. (In Defense of Interactive Graphics)



`Source` Russell Samora / The Pudding — *Responsive scrollytelling best practices*

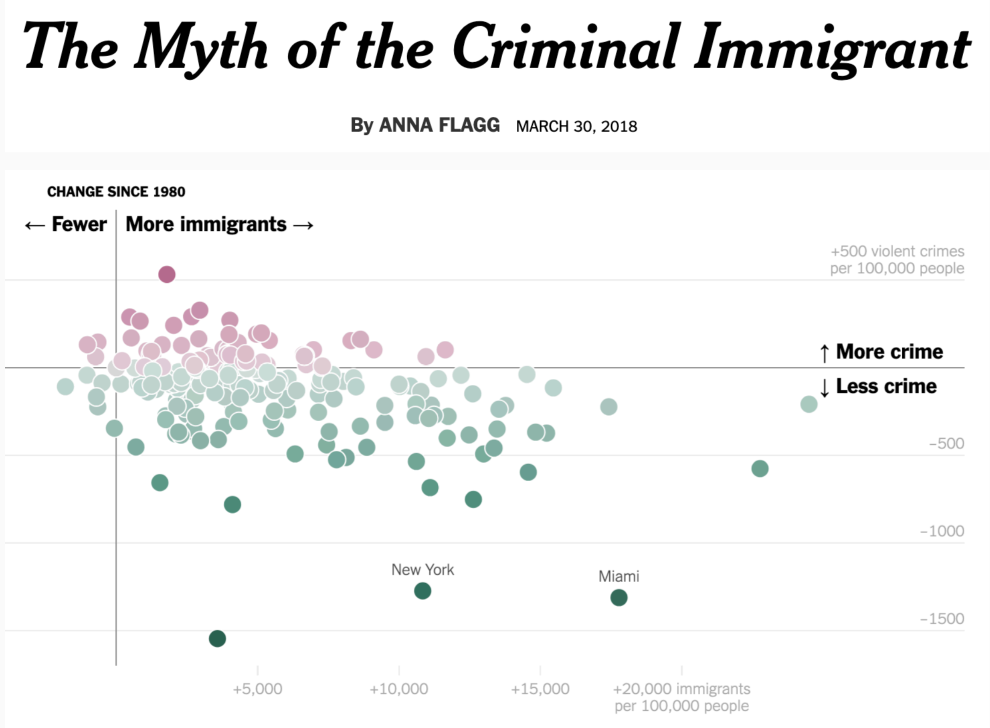
Default to static.

3 rules for visual storytelling

1. If you make the reader click or do anything other than scroll, something spectacular has to happen.
2. If you make a tooltip or rollover, assume no one will ever see it. If content is important for readers to see, don't hide it.
3. When deciding whether to make something interactive, remember that getting it to work on all platforms is expensive.

By Archie Tse, NYT, 2016

Case study



`Source` Anna Flagg for NYT The Upshot

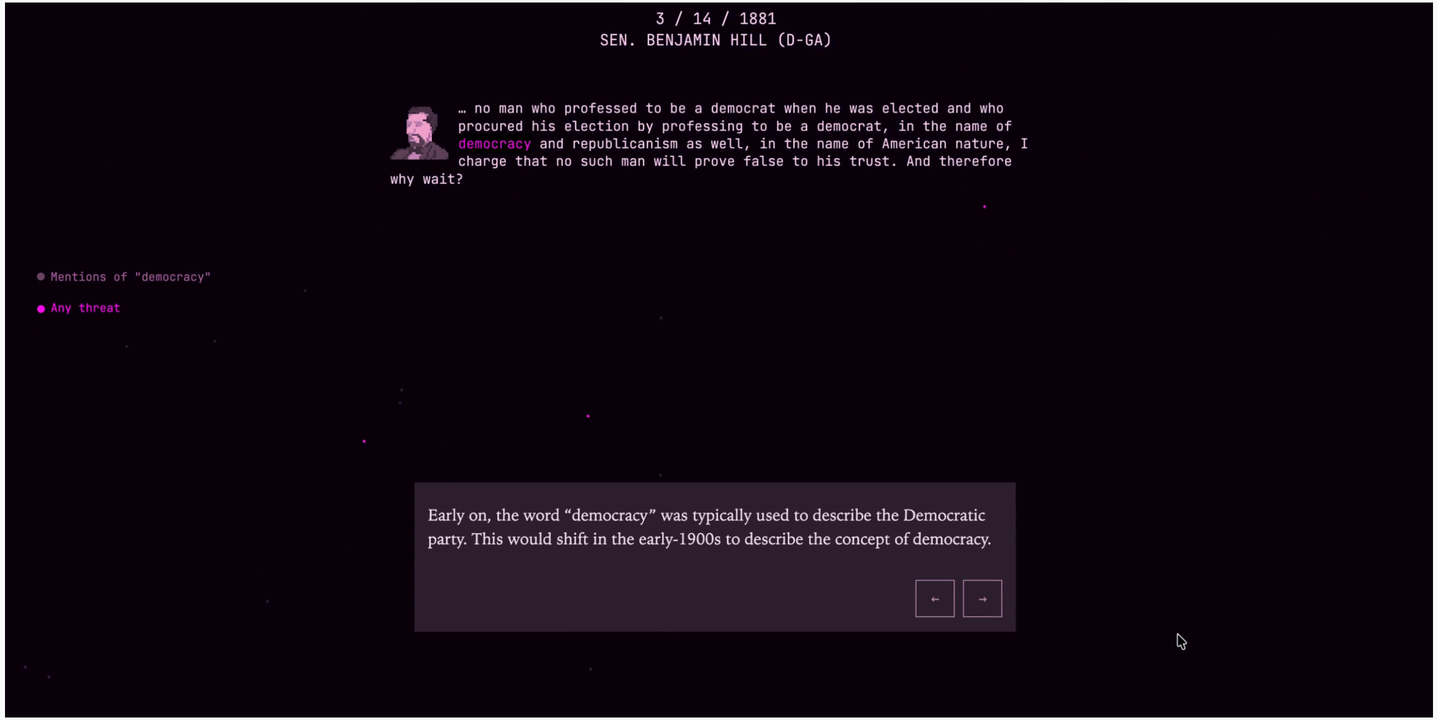
Scrollytelling is the right ceiling, often

Good scrollytelling

Scrolling is *also* an interaction — but it is the one your reader is already doing.

- one chart that **transforms** as you scroll,
- annotations that **arrive** at the right moment,
- never asks the reader to *do* anything except keep going.

This pattern carries the narrative without the cognitive overhead of filters, dropdowns, or dragging.



`Source` The Pudding — *In pursuit of democracy* (every use of the word "democracy" in the Congressional Record since 1880).

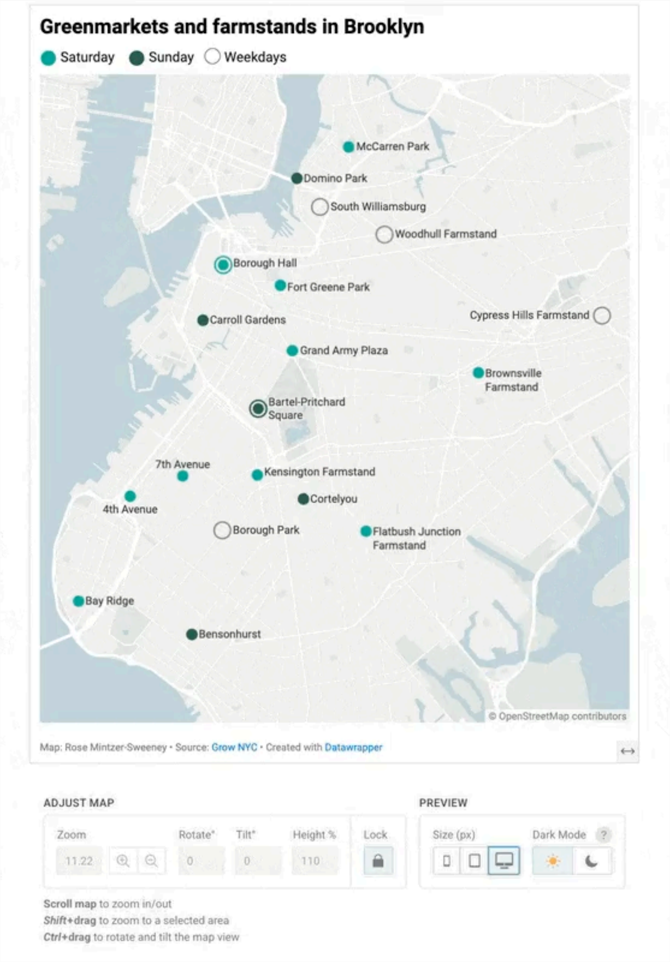
4. Build for the phone first

Where your readers actually are

Some evidence

- **Mobile is now the dominant news channel.** Reuters Institute *Digital News Report 2025* finds smartphones the most-used news device in most markets surveyed. (report)
- **In the US, 86%** of adults get news at least sometimes from a smartphone, tablet, or computer; **21%** prefer apps and websites over any other source. (Pew Research, *News Platform Fact Sheet*, 2025 — link)
- The mobile viewport width of common smartphones is **~400px wide**. Your typical reader sees the chart smaller than your laptop preview.

> A one-row, four-panel small-multiples chart that shines on a 27" monitor *dies* on a 390px column. Designing for desktop and "just hoping mobile works" is the modern equivalent of designing for print and assuming the website will be fine.



`Source` Datawrapper

Implications for your work

One chart, one finding.

In that sense, mobile forces a discipline on you. If you have three findings, you have three charts. Stop combining too much into one chart.

A mobile sanity check before you publish

1. Open the live page on your phone.
2. Read it with your phone held normally — not zoomed in.
3. Can you read every label? Can you tell what the headline finding is in 5 seconds?
4. If you are using ggplot, render at `width = 4, height = 5, dpi = 300` and look at it small.

Browser extensions like `Mobile simulator` let you emulate the mobile ezperience on desktop. If anything fails: cut a series, raise the type size, redo the title.

What "mobile-first" actually changes

Feature	Desktop	Mobile
Aspect ratio	landscape (16:9)	portrait or square
Series in one chart	up to ~6	1–2; rest go to small multiples or get cut
Type size	11–12pt	14–16pt minimum
Legend	OK	direct labels only
Title	one line	wraps to 2–3 lines
Axis labels	full text	abbreviated or removed
Small multiples	grid (e.g. 4×4)	1-column stack
Interactivity	hover tooltips	tap-and-hold; usually drop

Workshop II (10 min, in teams)

Workshop II — design a chart on the whiteboard

The brief

The **V-Dem Institute's 2024 release** shows the global electoral democracy index has fallen back to **1986 levels**. But the global average masks dramatic divergence: large democracies have backslid (US, India, Brazil under Bolsonaro), a handful of autocracies have liberalized, and entire regions move in opposite directions within the same decade.

You have **10 minutes**, in teams of 3–4, to **sketch one chart on the whiteboard** for a piece on this. No laptops.

Constraints

- **One chart only.**
- It can be **static or interactive.**

We'll walk around at the end.

The dataset — V-Dem v14, country-year panel

Variable	Type	Description
`country`	chr	~180 countries
`year`	int	1990 – 2023
`region`	chr	10 world regions
`v2x_polyarchy`	dbl	Electoral Democracy Index, 0–1
`regime_type`	chr	Closed autocracy → Liberal democracy (4 cats)
`population`	dbl	Total population, optional weight

Shape: ~180 countries × 34 years ≈ **6,100 country-year rows.**

`Source` V-Dem Institute, *Varieties of Democracy* · Democracy Report 2024

Your charts: live discussion

Your submitted charts

A few of you volunteered charts from your own data-bit and other works. We'll go through them one by one. For each, the room will:

1. **Read it cold.** What's the headline finding? (no help from the author)
2. **Spot the editorial choices.** What was foregrounded? What was cut?
3. **Provide constructive feedback.** Where does it shine? Where would we change it?
4. **Author's reply.** What did you mean to say, and did the chart say it?

The goal isn't to praise or pile on. It's to practise the act of *reading* a chart with the same care we apply to writing one.

Further reading

Books

- Claus Wilke, *Fundamentals of Data Visualization* — a standard reference.
- Alberto Cairo, *The Truthful Art* and *How Charts Lie* — both written for practitioners.
- Cole Nussbaumer Knaflic, *Storytelling with Data: Let's Practice!* — workshop-style, you're probably beyond this.

Blogs / Newsletters / Online material

- Datawrapper Blog — Lisa Charlotte Muth's *Data Vis Dispatch* is the single best weekly digest.
- Junk Charts — Kaiser Fung's redesigns.
- Open Visualization Academy — Free courses by Alberto Cairo and others.
- The Functional Art — Alberto Cairo.

Newsroom showcases

- FT Visual Vocabulary
- Reuters Graphics | Pudding | NYT Upshot
- Sigma Awards — annual data-journalism prize